



Self-adaptive competitive swarm optimizer: a memetic approach for global optimization and human-powered aircraft design

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Received: 17 January 2025 / Accepted: 6 June 2025

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Abstract

This paper presents a memetic approach of Competitive Swarm Optimizer (CSO) named Self-Adaptive Competitive Swarm Optimizer (SACSO). We focus on two key parameters of CSO: scaling factor and population size and introduce a parameter sorting scheme and a linear population reduction strategy to enhance the performance of SACSO. Specifically, the parameter sorting scheme assigns smaller scaling factors to better-performing particle individuals to encourage exploitative searchability, while larger scaling factors are allocated to poor-performing particle individuals to promote exploration in unknown search areas. The linear population reduction strategy adapts to different optimization phases, emphasizing exploration in the early phase and enhancing exploitation in the later phase. To investigate the performance of SACSO, we conducted comprehensive numerical experiments on CEC2017, CEC2022, and seven classic engineering problems. The experimental results and statistical analysis demonstrate the competitiveness of SACSO against six state-of-the-art optimizers and four advanced CSO variants. Ablation studies further validate the contribution of the two introduced strategies. Finally, we applied SACSO to real-world Human-Powered Aircraft (HPA) design problems, and the results highlight the potential of SACSO for solving various real-world optimization challenges. The source code of SACSO is available at <https://github.com/RuiZhong961230/SACSO>.

Keywords Competitive swarm optimizer (CSO) · Parameter sorting · Linear population reduction · Human-powered aircraft (HPA) design

1 Introduction

Evolutionary Computation (EC) has emerged as a highly efficient and versatile optimization tool over the past few decades, proving capable of addressing a wide range of complex and high-dimensional optimization problems across various fields, such as engineering, finance, logistics, and artificial intelligence [1–3]. Many nature-inspired approaches have been developed to tackle these intricate optimization challenges, with notable algorithms including Genetic Algorithm (GA), Differential Evolution (DE), and Particle Swarm Optimization (PSO). GA employs evolutionary concepts such as crossover and mutation to generate new offspring individuals and effectively search the solution space by mimicking natural selection and genetic inheritance [4, 5]. On the other hand, DE utilizes a differential mutation and crossover operator to enhance population diversity and guide the search process toward optimal or near-optimal solutions, which

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excels in handling continuous optimization problems [6, 7]. PSO draws inspiration from the collective behavior of birds and fish and balances exploration and exploitation by iteratively updating the velocity and position of particles in the search space based on individual and collective experiences [8–10].

Compared to traditional optimization techniques such as gradient-based methods, which rely heavily on gradient information and often struggle with non-differentiable, non-convex, or noisy functions [11, 12], Evolutionary Algorithms (EAs) offer several significant advantages. These include scalability, which allows EAs to handle a diverse range of problems from various optimization domains, from small-scale to large-scale and single-objective to multi-objective scenarios [13–15]. Additionally, EAs are easy to implement as they do not require detailed problem information like gradients or specific landscape features, which makes them suitable for black-box optimization problems where such details are unknown or difficult to compute. Another advantage of EAs is their robustness. EAs are less sensitive to problem-specific characteristics and are highly adaptable to dynamic and noisy environments, which is crucial for real-world applications where conditions often change unpredictably [16]. Furthermore, EAs are inherently parallelizable and well-suited for modern computational environments that leverage parallel and distributed computing frameworks, which significantly reduce computational time and enhance solution quality [17, 18]. These superior attributes make EAs powerful tools not only for theoretical research but also for practical applications in real-world optimization challenges [19, 20].

Among the diverse range of EAs, the Competitive Swarm Optimizer (CSO) stands out as a notable advancement, integrating a competitive mechanism into the efficient framework of PSO. Originally proposed by Cheng and Jin in 2014 [21], CSO combines the exploration strengths of PSO with a unique competition-based selection process to improve search efficiency. A detailed introduction to CSO is provided in Sect. 2.1. As a well-designed and robust optimizer, CSO has been effectively extended to address a variety of optimization challenges, including single-objective problems [22, 23], multi-objective problems [24, 25], large-scale optimization tasks [26, 27], and computationally expensive optimization scenarios [28, 29]. These extensions showcase CSO's remarkable scalability and adaptability across different optimization domains. However, despite its success, CSO still faces several challenges. One of the primary issues is achieving a balance between the exploration and exploitation phases of the search process, which is crucial for avoiding premature convergence to suboptimal solutions. Another limitation is the "two steps forward, one step backward" phenomenon, where the optimization process is occasionally hindered by setbacks that slow down progress toward optimal solutions [30]. These challenges

highlight the necessity for further enhancements to CSO to improve its performance for specific problem domains. We attribute these limitations of CSO to its simple architecture and single search pattern, which can be alleviated by integrating intelligent and automatic parameter adaptation schemes. Therefore, this paper proposes a novel memetic approach of CSO termed Self-Adaptive CSO (SACSO), where two parameter adaptation schemes are introduced into CSO: the parameter sorting scheme and the linear population reduction strategy. The integration of these two schemes significantly accelerates the optimization convergence and enhances the scalability of SACSO. To comprehensively investigate the performance of SACSO, we conduct numerical experiments in CEC2017, CEC2022 benchmarks, and seven classic engineering optimization problems. Comparative experiments against state-of-the-art optimizers and advanced variants of CSO demonstrate the competitiveness and domination of the proposed SACSO. Furthermore, we extend our proposed SACSO to address the human-powered aircraft (HPA) design challenges. The experimental results and statistical analysis confirm that SACSO has great potential in solving real-world engineering applications. Briefly, the contributions of this paper as summarized as follows.

- We introduce the parameter sorting scheme and the linear population reduction strategy to CSO and proposed SACSO.
- We conduct comprehensive numerical experiments in CEC2017, CEC2022 benchmarks, and classic engineering problems to investigate the performance of SACSO against state-of-the-art optimizers.
- We further apply the proposed SACSO to solve human-powered aircraft (HPA) design tasks.
- The ablation experiments are conducted to investigate the components in the proposed SACSO independently.

The remainder of this paper is structured as follows: Sect. 2 presents the related works. Section 3 details the proposed SACSO. The numerical experiments are introduced in Sect. 4. Finally, Sect. 5 concludes this paper.

2 Related works

2.1 Particle swarm optimization (PSO) and competitive swarm optimizer (CSO)

PSO is a population-based stochastic optimization that simulates the behaviors of bird flocks [31]. After the population

initialization within the search domain, PSO iterates Eq. (1) to refine the population and approach optima.

$$\begin{aligned} \mathbf{v}_i^{t+1} &= \omega \cdot \mathbf{v}_i^t + c_1 \cdot \mathbf{r}_1 \cdot (\mathbf{x}_{pbest}^t - \mathbf{x}_i^t) + c_2 \cdot \mathbf{r}_2 \cdot (\mathbf{x}_{gbest}^t - \mathbf{x}_i^t) \\ \mathbf{x}_i^{t+1} &= \mathbf{x}_i^t + \mathbf{v}_i^{t+1} \end{aligned} \quad (1)$$

where $\mathbf{v}_i^t$ and $\mathbf{x}_i^t$ denote the velocity and position of the i^{th} particle in the t^{th} iteration, respectively. ω is the inertia weight, c_1 and c_2 indicate acceleration coefficients, $\mathbf{r}_1$ and $\mathbf{r}_2$ are two random vectors where each element is in $[0, 1]$. $\mathbf{x}_{pbest}^t$ denotes the cognitive coefficient, which is the optimum of $\mathbf{x}_i$ found so far, while $\mathbf{x}_{gbest}^t$ denotes social coefficient, which is the optimum of the population [32].

Motivated by the efficient framework of PSO, Cheng and Jin further introduced a competitive mechanism to PSO and proposed CSO [21]. A demonstration of CSO is shown in Fig. 1.

In each iteration, CSO randomly selects two individual $\mathbf{x}_{r1}^t$ and $\mathbf{x}_{r2}^t$ for competition. The winner with a better fitness value is denoted as $\mathbf{x}_w^t$ and the loser is denoted as $\mathbf{x}_l^t$. After the entire population is well divided into the winner or the loser, the winner individuals directly survive to the next iteration while losers learn the knowledge from winners and the centroid of the population using Eq. (2).

$$\begin{aligned} \mathbf{v}_l^{t+1} &= \mathbf{r}_1 \cdot \mathbf{v}_l^t + \mathbf{r}_2 \cdot (\mathbf{x}_w^t - \mathbf{x}_l^t) + \varphi \cdot \mathbf{r}_3 \cdot (\mathbf{x}_{mean}^t - \mathbf{x}_l^t) \\ \mathbf{x}_l^{t+1} &= \mathbf{x}_l^t + \mathbf{v}_l^{t+1} \end{aligned} \quad (2)$$

where r_1 , r_2 , and r_3 are three random vectors where each element is in $[0, 1]$, φ is a scaling parameter to control the learning phase ($\mathbf{x}_{mean}^t - \mathbf{x}_l^t$), and $\mathbf{x}_{mean}^t$ is the centroid of the population. In each iteration, only half of the population is partitioned as losers and subsequently updated. In summary, the pseudocode of CSO is presented in Algorithm 1.

The effectiveness of CSO has been mathematically proven to ensure global convergence and has been practically validated as an efficient approach across various optimization domains [21].

2.2 Literature review for CSO

As an efficient and effective optimization approach, CSO has attracted widespread attention from researchers and scholars. Yang et al. [33] proposed a novel multiple predominant particles guide learning mechanism and incorporated it into CSO to develop the Segment-based Predominant Learning Swarm Optimizer (SPLSO). Motivated by the cooperative coevolution (CC) framework, SPLSO first randomly divides decision variables into several segments. Variables in different segments evolve by learning from different exemplars,

Algorithm 1 CSO

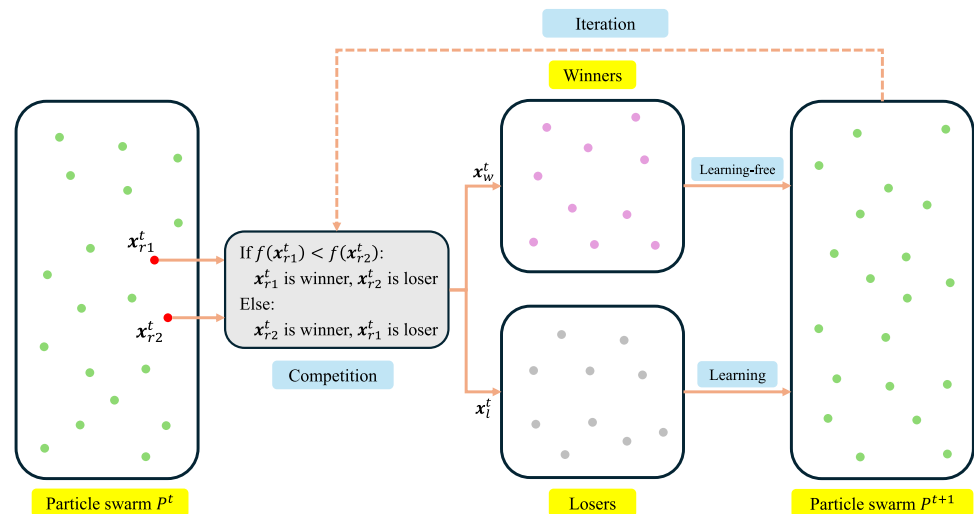
Require: Swarm size: N , Dimension: D , Maximum iteration: T
Ensure: Optimum: $\mathbf{x}_{best}^t$

- 1: Initialize the particle swarm X
- 2: $t = 0$
- 3: Record the current optimum $\mathbf{x}_{best}^t$
- 4: **while** $t < T$ **do**
- 5: $P \leftarrow \text{copy}(X)$
- 6: **while** P is not empty **do**
- 7: Select two random particle individuals $\mathbf{x}_{r1}^t$ and $\mathbf{x}_{r2}^t$
- 8: **if** $f(\mathbf{x}_{r1}^t)$ is better than $f(\mathbf{x}_{r2}^t)$ **then**
- 9: Assign $\mathbf{x}_{r1}^t$ as $\mathbf{x}_w^t$ and $\mathbf{x}_{r2}^t$ as $\mathbf{x}_l^t$
- 10: **else**
- 11: Assign $\mathbf{x}_{r1}^t$ as $\mathbf{x}_l^t$ and $\mathbf{x}_{r2}^t$ as $\mathbf{x}_w^t$
- 12: **end if**
- 13: Remove $\mathbf{x}_{r1}^t$ and $\mathbf{x}_{r2}^t$ from P
- 14: Survive $\mathbf{x}_w^t$ to X^{t+1}
- 15: State Update $\mathbf{x}_l^t$ using Eq. (2) and survive to X^{t+1}
- 16: **end while**
- 17: Record the current optimum $\mathbf{x}_{best}^t$
- 18: $t \leftarrow t + 1$
- 19: **end while**
- 20: **return** $\mathbf{x}_{best}^t$

while variables within the same segment evolve by learning from the same exemplar. Second, a novel predominant learning scheme enables multiple predominant individuals to guide the optimization direction for each individual, with each predominant particle responsible for one segment of dimensions. The integration of these two competitive strategies in SPLSO accelerates optimization convergence and enhances accuracy.

Inspired by the teaching and learning behaviors in pedagogy, where teachers often divide a mixed-level group of students into different levels based on their cognitive and learning abilities and then apply tailored teaching activities, Yang et al. [34] proposed a Level-based Learning Swarm Optimizer (LLSO) for handling complex optimization challenges. Initially, a level-based learning mechanism is integrated to divide particle individuals into different levels based on their fitness values and apply distinct learning strategies to construct offspring independently at each level. Second, a new exemplar selection mechanism is developed to select two predominant particles from two different higher levels to guide the learning phase of other particles. Practical experiments and theoretical analysis confirm the effectiveness of embedding these two strategies into CSO.

Lan et al. [35] presented a Two-Phase Learning-based Swarm Optimizer (TPLSO), which integrates the cooperative learning behavior in human society into CSO. The proposed TPLSO consists of two phases: comprehensive learning and elite learning. In the comprehensive learning phase, TPLSO randomly selects three unique individuals to form a group and utilizes the competitive mechanism to construct the offspring individuals. After the whole particle population finishes the comprehensive learning phase, TPLSO sorts the particle pop-

Fig. 1 A demonstration of CSO

ulation based on the fitness value and subsequently enters the elite learning phase to exploit the promising search domains further.

Pakarat et al. [36] presented an Opposition-based Modified CSO (OMCSO) and extended it to address the Wireless Sensor Network (WSN) configuration problems. The Opposition-based Learning scheme is responsible for updating the winner particle individuals to improve the probabilities of finding the optimum and accelerate the optimization convergence, while the hyperparameter φ is ignored in the learning mechanism for losers to enhance the scalability of OMCSO. Simulation experiments in maximizing the coverage rate and minimizing the energy consumption of WSN configuration confirm the feasibility and competitiveness of OMCSO.

Huang et al. [37] introduced a novel enhanced CSO named DMCACSO, which is integrated with dynamic multi-competition mechanisms and a convergence acceleration strategy for large-scale global optimization. DMCACSO adopts an improved multi-round competition scheme to more effectively identify high-quality winners and losers within the population. To balance exploration and exploitation, the number of competition rounds is dynamically adjusted using the well-established 1/5 success rule to enhance population diversity and promote the generation of higher-quality solutions. To further prevent premature convergence and improve the ability to escape local optima, a hybrid evolutionary strategy is integrated, which activates when stagnation is detected. Specifically, loser particles adaptively learn from the winners and the mean position of the swarm or the top-performing individuals to foster both convergence and diversity. Comprehensive experimental evaluations on the 1000-dimensional CEC2010 and CEC2013 benchmark suites demonstrate that the proposed DMCACSO significantly outperforms 11 state-of-the-art optimization methods.

Furthermore, the application to feature selection tasks confirms the practical effectiveness of DMCACSO in achieving superior classification performance compared to existing techniques.

Focusing on the fixed-charge transportation (FCT) problem, Chauhan and Shivani [38] introduced an offline learning-based CSO (OLCSO) to handle this NP-hard problem, which leverages historical data and collective intelligence to enhance solution quality. Inspired by offline learning principles, candidate solutions in OLCSO learn from high-performing peers rather than relying solely on real-time feedback. Furthermore, the ranking-based peer learning mechanism adaptively adjusts the learning intensity depending on the rank difference to ensure that lower-ranked particles adapt more aggressively toward better solutions. Moreover, a two-phase cooperative evolutionary mechanism and an adaptive mutation strategy maintain population diversity during optimization and balance the trade-off between exploration and exploitation. The performance of OLCSO is rigorously evaluated against state-of-the-art optimizers and multiple performance metrics, and the results confirm the effectiveness of the proposed OLCSO.

The aforementioned variants of CSO primarily focus on single-objective optimization. Meanwhile, CSO has also been actively explored in the multi-objective optimization domain. Zhang et al. [39] proposed a Competitive Mechanism-based Multi-Objective PSO (CMOPSO), which inherits the efficient framework of the Multi-Objective PSO (MOPSO). Different from many Multi-Objective EAs (MOEAs) that require a computationally expensive external archive, CMOPSO does not rely on historical knowledge from the evolutionary path, which reduces computational costs during optimization. Additionally, the competitive mechanism from CSO is adapted to the multi-objective domain, where the concept of Pareto front domination

replaces the simple fitness comparison. Experimental results demonstrate the superiority of CMOPSO in achieving high-quality solutions and faster convergence.

Liu et al. [40] introduced a specialized Comprehensive Competitive Learning approach for CSO (CCLCSO), which incorporates three competition mechanisms to guide particle searches. In this approach, the original competition mechanism from CSO distinguishes winner and loser individuals within the particle swarm, while cognitive and social competition mechanisms each select a winning particle to serve as the cognitive and social components, respectively. This unique and efficient scheme guides the search for the loser particles. Experimental results and statistical analysis confirm the searchability and scalability of CCLCSO in multi-objective optimization, which enhances the diversity of search preferences.

Inspired by the concept of the flipped classroom, which shifts the focus from direct instruction to enhancing students' capacity for autonomous learning, Qi et al. [41] proposed the Self-exploratory Competitive Swarm Optimizer (SECSO) for Large-scale Multi-objective Optimization Problems (LSMOs). SECSO is advantageous due to its hyperparameter-free characteristic and simple motivation. In SECSO, particles learn from their neighbors and other particles, which accelerates their evolution while maintaining swarm diversity and enhancing optimization convergence. Experimental results demonstrate that SECSO can find a set of non-dominated solutions with only an affordable computational cost.

Gao et al. [42] proposed LSMCSO-SS, an enhanced variant of the CSO tailored for large-scale multi-objective optimization problems (LSMOPs), which integrates a space sampling strategy to generate a set of guiding individuals that are uniformly distributed across the decision space to improve population diversity and facilitate a more informed exploration of the search space. Furthermore, a novel velocity update component is introduced for the loser particles, which leverages dominance relationships to direct them toward superior solutions. Experimental evaluations on the LSMOP benchmark suite with up to 5000 decision variables demonstrate that LSMCSO-SS consistently outperforms seven state-of-the-art algorithms on most test instances, as assessed by the Inverted Generational Distance Plus (IGD+) metric. Ablation studies further validate the individual contributions of both the space sampling strategy and the newly introduced velocity component, confirming their critical roles in achieving the observed performance gains.

More detailed reviews of CSO can be found in [43, 44].

2.3 Human-powered aircraft (HPA) design

The human-powered aircraft (HPA) design is an important challenge in the broader field of aeronautics and structural



Fig. 2 A demonstration of human-powered aircraft [45]

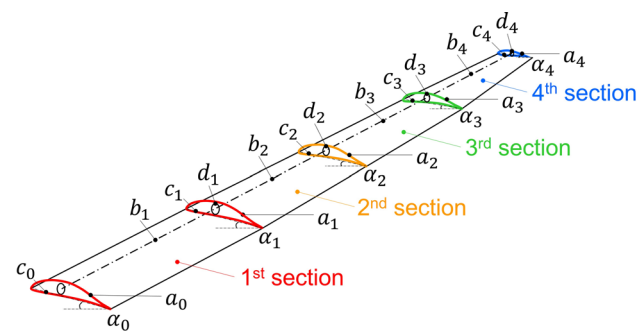


Fig. 3 Defined variables in the wing shape [45]

engineering that intersects significantly with optimization, particularly in lightweight structures, aerodynamic efficiency, and human endurance. The primary challenge in HPA is to design an aircraft that can be propelled by human power alone, which requires optimizing various to minimize weight and drag while maximizing lift and power efficiency. Figure 2 presents a demonstration of human-powered aircraft.

Wing-shape optimization plays a crucial role in HPA design. Here, Nobuo Namura [45] separates the right wing into n segments to form $(n + 1)$ cross-sections. Figure 3 demonstrates an instance when $n = 4$, and real-coded variables are as follows.

- a_i : the airfoil selection.
- b_i : the segment length.
- c_i : the chord length.
- α_i : the angle of attack.
- d_i : the diameter of the CFRP pipe forming the main spar.

The optimization problems defined in HPA design are formulated in Eq. (3).

$$\text{HPA101} : f(\mathbf{x}) = D(\mathbf{x})$$

$$\text{HPA102} : f(\mathbf{x}) = P(\mathbf{x}) = D(\mathbf{x})V(\mathbf{x})/\eta$$

$$\text{HPA103} : f(\mathbf{x}) = -V(\mathbf{x})$$

$$\text{HPA131} : f(\mathbf{x}) = D(\mathbf{x})$$

$$g_1(\mathbf{x}) = n_m n_s \epsilon_{\max}(\mathbf{x})/\epsilon_u - 1$$

$$g_2(\mathbf{x}) = B(\sin \gamma(\mathbf{x}) - \sin \gamma_u)/2$$

$$g_3(\mathbf{x}) = -\delta_{\text{park}}(\mathbf{x})$$

$$\text{HPA142} : f(\mathbf{x}) = P(\mathbf{x}) = D(\mathbf{x})V(\mathbf{x})/\eta$$

$$g_1(\mathbf{x}) = n_m n_s \epsilon_{\max}(\mathbf{x})/\epsilon_u - 1$$

$$g_2(\mathbf{x}) = 1 - (V(\mathbf{x})/V_{\min})^3$$

$$\text{HPA143} : f(\mathbf{x}) = -V(\mathbf{x})$$

$$g_1(\mathbf{x}) = n_m n_s \epsilon_{\max}(\mathbf{x})/\epsilon_u - 1$$

$$g_2(\mathbf{x}) = B(\sin \gamma(\mathbf{x}) - \sin \gamma_u)/2$$

$$g_3(\mathbf{x}) = -\delta_{\text{park}}(\mathbf{x})$$

$$g_4(\mathbf{x}) = P(\mathbf{x}) - P_{\max} \quad (3)$$

Here, HPA131, HPA142, and HPA143 are constrained versions of HPA101, HPA102, and HPA103, respectively. The functions $P(\cdot)$, $D(\cdot)$, $V(\cdot)$, $E(\cdot)$, and $B(\cdot)$ represent the required power, drag, cruise speed, wing efficiency, and wingspan, respectively. The decision variables δ and δ_{park} denote the wingtip displacement (positive upwards) during flight and when parked, respectively, relative to the height of the wing root. $\epsilon_{\max}$ indicates the maximum strain in CFRP pipes under both flight and parked conditions, while γ represents the dihedral angle at the wingtip during flight. Additionally, the constants in the mathematical model are summarized as follows:

- $\eta = 0.85 \times 0.95$: the product of propeller and drive-train efficiency.
- $n_m = 1.5$, $n_s = 2$, and $\epsilon_u = 0.0027$: the load factor, the safety factor, and the reference strain.
- $\gamma_u = 8$: the maximum dihedral angle.
- $P_{\max} = 400$: the maximum required power.
- $V_{\min} = 7.3$: the minimum cruise speed.

More details of the HPA benchmark can be referred to [45].

3 Our proposal: SACSO

This section introduces the proposed SACSO. Figure 4 presents the flowchart of SACSO, with the main modifications highlighted in red. SACSO inherits the main

architecture of CSO and incorporates two efficient parameter adjustment strategies: a parameter sorting scheme and a linear population reduction strategy. These strategies are described in detail in the following sections.

3.1 Parameter sorting scheme

The parameter sorting scheme proposed by Zhou et al. [46] was initially integrated into JADE. In each iteration, JADE generates a crossover rate vector $CR = [Cr_1, Cr_2, \dots, Cr_N]$, where the length of CR corresponds to the population size. This vector is then sorted based on the fitness of the population in JADE. Specifically, better-performing parent individuals are assigned smaller crossover rates to produce offspring, while poorer-performing parents are assigned larger crossover rates. The motivation behind this scheme is that better-performing parent individuals with smaller crossover rates can transfer more valuable information to the next iteration. Conversely, poorer-performing parents, which receive larger crossover rates, can probabilistically transfer useful genetic material while discarding ineffective information.

Motivated by the efficiency and simplicity of this scheme, we extend the parameter sorting scheme to CSO. A demonstration is provided in Fig. 5. Suppose there are six particle individuals in the fitness landscape, as shown in Fig. 5a. The parameter sorting scheme in CSO then samples a 6-dimensional φ vector from a uniform distribution $U(0.001, 0.5)$ as recommended in [21]. Also, as confirmed in [21], setting $\varphi > 0$ ensures the global convergence of CSO. Subsequently, the sorting operator arranges the particle individuals and the φ vector, and the competitive mechanism in SACSO randomly selects two particle individuals, as shown in Fig. 5d. The better-performing individual is denoted as $\mathbf{x}_w^t$ and the worse-performing one as $\mathbf{x}_l^t$. The better individual $\mathbf{x}_w^t$ directly survives to the next iteration, while the loser $\mathbf{x}_l^t$ updates using Eq. (2). This selection-competition process is repeated until all particle individuals have been selected.

Similar to the mechanism in [46], when a relatively well-performing particle individual is denoted as the loser, a small φ value allows for an exploitative search to find high-quality solutions. This is also consistent with the proximate optimality principle (POP) [47], which suggests that an exploitative search around elite individuals is more likely to yield high-quality solutions. Conversely, if a poorly performing particle is selected as the loser, a large φ value enables SACSO to explore unknown search areas. This simple yet automatic parameter adaptation scheme is effective and promising for accelerating optimization convergence while avoiding local optima.

Fig. 4 The flowchart of SACSO

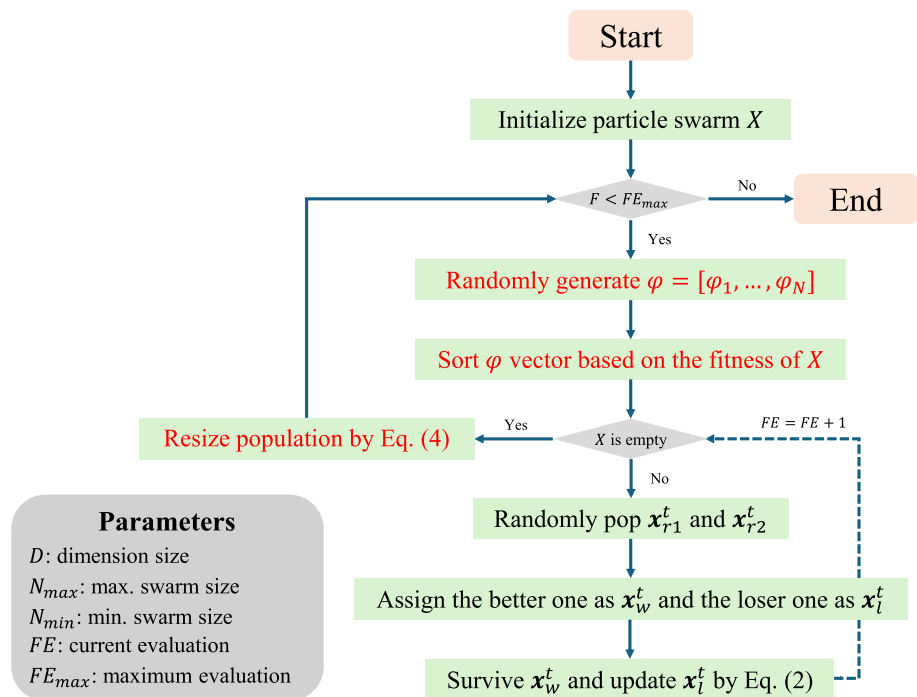
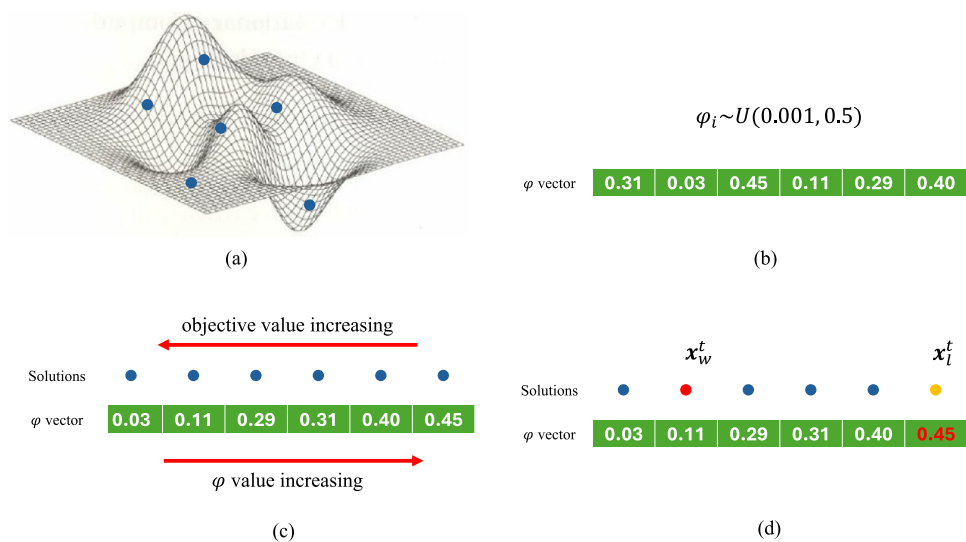


Fig. 5 A demonstration of parameter sorting scheme in SACSO within six particle individuals. **a** The distribution of six particle individuals in the fitness landscape. **b** Sampling φ vector from a uniform distribution $U(0.001, 0.5)$. **c** Sorting the particle individuals and the φ vector. **d** Randomly selecting two particle individuals to employ the competitive mechanism in CSO



3.2 Linear population reduction strategy

The population reduction strategy proposed in [48, 49] is an efficient approach to balance exploration and exploitation during optimization. The core idea of this strategy is to adjust the population size throughout different phases of the optimization. In the early phase, the algorithm maintains a large population size, which enhances explorative search capabilities. As the evolution progresses to the middle phase, a moderate population size allows for a balanced search between exploration and exploitation. Finally, in the late phase, a smaller population size focuses on exploitative search, which contributes to accelerating convergence

and improving accuracy. This expert-designed methodology enables the algorithm to automatically adjust the optimization preference according to different phases of the optimization process.

Success-History Adaptive DE with Linear Population Reduction (L-SHADE) [50], one of the most renowned DE variants, is well-known for its high performance in CEC benchmark competitions. Therefore, we integrate the deterministic linear population reduction strategy from L-SHADE into SACSO, as formulated in Eq. (4).

$$N^t = \text{round} \left(\frac{N_{min} - N_{max}}{FE_{max}} \cdot FE + N_{max} \right) \quad (4)$$

where N_{max} and N_{min} are the maximum and minimum population sizes, respectively. FE and FE_{max} are the current and maximum fitness evaluations, respectively. The integration of the linear population reduction strategy is expected to further improve the optimization performance of SACSO.

In summary, the pseudocode of SACSO is presented in Algorithm 2.

Algorithm 2 SACSO

Require: Max. and Min swarm size: N_{max} and N_{min} , Dimension: D ,
Maximum fitness evaluations: FE_{max}

Ensure: Optimum: $\mathbf{x}_{best}^t$

```

1: Initialize the particle swarm  $X$ 
2:  $FE = 0$ 
3: Record the current optimum  $\mathbf{x}_{best}^t$ 
4: while  $t < T$  do
5:    $P \leftarrow \text{copy}(X)$ 
6:   Construct the  $\varphi$  vector with a uniform distribution
7:   Sort  $\varphi$  vector in ascending order
8:   while  $P$  is not empty do
9:     Select two random particle individuals  $\mathbf{x}_{r1}^t$  and  $\mathbf{x}_{r2}^t$ 
10:    if  $f(\mathbf{x}_{r1}^t)$  is better than  $f(\mathbf{x}_{r2}^t)$  then
11:      Assign  $\mathbf{x}_{r1}^t$  as  $\mathbf{x}_w^t$  and  $\mathbf{x}_{r2}^t$  as  $\mathbf{x}_l^t$ 
12:    else
13:      Assign  $\mathbf{x}_{r1}^t$  as  $\mathbf{x}_l^t$  and  $\mathbf{x}_{r2}^t$  as  $\mathbf{x}_w^t$ 
14:    end if
15:    Remove  $\mathbf{x}_{r1}^t$  and  $\mathbf{x}_{r2}^t$  from  $P$ 
16:    Survive  $\mathbf{x}_w^t$  to  $X^{t+1}$  and save  $\mathbf{x}_l^t$  to  $X_l^t$ 
17:  end while
18:  Sort  $X_l^t$  based on fitness value in ascending order
19:  for  $i = 0$  to  $N/2$  do
20:    Update  $X_{l,i}^t$  using Eq. (2) with  $\varphi_i$  and survive to  $X^{t+1}$ 
21:  end for
22:  Resize the particle swarm  $X$  using Eq. (4)
23:  Record the current optimum  $\mathbf{x}_{best}^t$ 
24:   $t \leftarrow t + 1$ 
25: end while
26: return  $\mathbf{x}_{best}^t$ 

```

3.3 Computational complexity analysis

This section analyzes the computational complexity of SACSO. Supposing the population size is N , the maximum iteration is T , and the dimension size of the problem is D . The computational complexity of components in SACSO is summarized as follows.

- Particle swarm initialization: $O(N \times D)$
- φ vector construction: $O(N)$
- φ vector and swarm arrangement: $O(2N \times \log N)$
- The poor-performing individuals update: $O(N \times D)$
- Particle swarm resizing: $O(1)$

In summary, the computational complexity of SACSO is $O(N \times D + T \times (N + 2N \times \log N + N \times D + 1)) := O(T \times N \times (\log N + D))$. Since the sorting operator is

involved in SACSO, it consumes an additional computational budget than CSO. However, considering the significant improvement from the original CSO, this extra computational cost is acceptable.

4 Numerical experiments

In this section, we conduct comprehensive numerical experiments and rigorous statistical analysis to investigate the performance of our proposed SACSO in various optimization domains. Initially, Sect. 4.1 introduces the settings of experiments including experimental environments and implementations, benchmark functions, and competitor algorithms. The detailed experimental results are summarized in Sect. 4.2.

4.1 Experimental settings

4.1.1 Experimental environments and implementations

We conduct numerical experiments in the following platform:

- Device: ThinkPad T14s Gen 6 Snapdragon (14)
- OS: Windows 11 Pro 64 ARM
- CPU: Snapdragon X Elite X1E-78-100 Processor (3.40 GHz)
- GPU: Integrated Qualcomm Adreno GPU
- RAM: 32 GB LPDDR 5X-8448MHz (Soldered)
- HDD: 1 TB SSD M.2 2242 PCIe Gen4 TLC Opal
- Language: Python 3.11

The detailed introduction ensures the reproducibility of this research.

4.1.2 Benchmark functions

The following benchmarks serve as the test-bed to investigate the performance of SACSO.

- IEEE-CEC2017: CEC2017 contains 29 various benchmark functions. The detailed information is summarized in Appendix A Table 18. Notice that f_2 is discarded and f_{30} is as the compensation.
- IEEE-CEC2022: CEC2022 contains 12 various benchmark functions. The detailed information is summarized in Appendix A Table 19.
- Engineering problems: We apply SACSO to optimize the Cantilever Beam Design Problem (CBDP), Corrugated Bulkhead Design Problem (CBHDP), Gear Train Design Problem (GTDP), Piston Lever Design Problem (PLDP), Speed Reducer Design Problem (SRDP), Three Bar Truss

Table 1 Basic information of seven engineering problems

Name	Abbr.	Dim.	# of constraints
Cantilever beam design problem	CBDP	5	1
Corrugated bulkhead design problem	CBHDP	4	6
Gear train design problem	GTDP	4	0
Piston lever design problem	PLDP	4	4
Speed reducer design problem	SRDP	7	11
Three bar truss design problem	TBTDP	2	3
Tubular column design problem	TCDP	2	6

Design Problem (TBTDP), and Tubular Column Design Problem (TCDP). The basic information of these problems is summarized in Table 1.

- HPA problems: HPA problems in Sect. 2.3 are employed as the testbed.

CEC benchmarks are available in the Opfunu library [51], the engineering suite is provided by the Enoppy library [52], and the HPA problems are obtained from [45].

4.1.3 Competitor algorithms and parameters

This section introduces the competitor algorithms and parameter settings. To comprehensively and fairly investigate the performance of L-SHACSO, we adopt two categories of MAs for comparison, which are listed as follows:

- State-of-the-art optimizers: L-SHADE [50], Improved L-SHADE (iL-SHADE) [53], Multi-Population Ensemble DE (MPEDE) [54], jSO [55], L-SHADE-cnEpSin [56], and DEA^2H^2 [57].
- Efficient variants of CSO: CSO [21], Modified CSO (MCSO) [58], Orthogonal Learning CSO (OLCSO) [59], and Three-Phase CSO (TPCSO) [60].

The advanced DE variants are open access at the PyADE library,¹ and the parameter settings of optimizers are summarized in Table 2.

The parameters listed in Table 2 are the optimal configurations suggested in corresponding papers for a fair comparison. The maximum fitness evaluations (FEs) are fixed at $1000 \times D$ (D = dimension size) for CEC2017 and CEC2022, 10,000 for the engineering problems, and 1000 for HPA design. The trial run is fixed at 30 to minimize the influence of randomness. Given that engineering and HPA problems contain constraints, a static penalty function [61] defined in Eq. (5) is equipped with optimizers to handle con-

Table 2 Detailed parameter settings of optimizers

MAs	Parameters	Value
L-SHADE	N_{max} and N_{min}	$18 \times D$ and 4
	μ_F and μ_{Cr}	0.5 and 0.5
	σ_F and σ_{Cr}	0.1 and 0.1
CSO	N	200
	φ	0.15
iL-SHADE	N_{max} and N_{min}	$12 \times D$ and 4
	μ_F and μ_{Cr}	0.5 and 0.8
	σ_F and σ_{Cr}	0.1 and 0.1
MPEDE	N_{max} and N_{min}	250 and 20
	μ_F and μ_{Cr}	0.5 and 0.5
	σ_F and σ_{Cr}	0.1 and 0.1
	λ_1, λ_2 , and λ_3	0.2, 0.2, and 0.2
jSO	N_{max} and N_{min}	$12 \times D$ and 4
	p_{max} and p_{min}	0.25 and 0.125
	M_F	0.3
L-SHADE-cnEpSin	N_{max} and N_{min}	$18 \times D$ and 4
	μ_F and μ_{Cr}	0.5 and 0.5
	σ_F and σ_{Cr}	0.1 and 0.1
	memory size H	5
MCSO	ps and pc	0.5 and 0.4
	N	200
	$\varphi_{1,max}$ and $\varphi_{1,min}$	0.3 and 0
OLCSO	$\varphi_{2,max}$ and $\varphi_{2,min}$	0.3 and 0
	N	200
TPCSO	ζ	0.001
	N	200
DEA^2H^2	φ_1 and φ_2	0.3 and 0.3
	N	100
SACSO	R	1
	N_{max} and N_{min}	400 and 4
	φ_{max} and φ_{min}	0.5 and 0.001

¹ <https://github.com/xKuZz/pyade>.

straints.

$$F(\mathbf{x}_i) = f(\mathbf{x}_i) + w \cdot \sum_{i=1}^m (\max(0, g_i(\mathbf{x}_i))) \quad (5)$$

where $F(\cdot)$ is the fitness function, $f(\cdot)$ is the objective function, and $g_i(\cdot)$ is the constraint function. w is a penalty constant to control the impact of constraints violation, which is fixed at $10e7$ by default [62, 63].

4.2 Experimental results and statistical analysis

4.2.1 Experiments on CEC2017

The detailed experimental results and statistical analysis are presented in Appendix B. To identify the statistical analysis between SACSO and other competitors, we first adopt the Mann–Whitney U test between every pair of algorithms, and the Holm multiple comparison test is subsequently employed to correct the p-value obtained from the previous Mann–Whitney U test. "+", "≈", and "−" indicate that the proposed SACSO is significantly better, without statistical significance, and significantly worse than the certain competitor algorithm. Table 3 summarizes the results of statistical analysis, average ranks of optimizers, and corrected average ranks in CEC2017. Figures 6 and 7 present the convergence curves, boxplots, and the proportion variation between exploration and exploitation in CEC2017 representative functions.

The CEC2017 benchmark suite comprises 29 functions, ranging from unimodal to multimodal, hybrid, and composite, with increasing complexity in their fitness landscapes. In general, SACSO achieved average ranks of 3.4 in 30-D and 3.7 in 50-D, ranking as the best and second-best, respectively. When compared to state-of-the-art DE variants such as L-SHADE, iL-SHADE, jSO, and L-SHADE-cnEpSin, as well as efficient CSO implementations like MCSO, OLC SO, and TPCSO, SACSO demonstrates significant competitiveness and remarkable optimization performance.

Furthermore, the comparison between advanced CSO variants and SACSO demonstrates the effectiveness of integrating the parameter sorting scheme and linear population reduction strategy. This integration significantly accelerates optimization convergence and achieves competitive performance compared to advanced CSO variants.

4.2.2 Experiments on CEC2022

The detailed experimental results and statistical analysis are presented in Appendix C. Table 4 summarizes the results of statistical analysis, average ranks of optimizers, and corrected average ranks in CEC2022. Figures 8 and 9 present the convergence curves, boxplots, and the proportion variation

Table 3 The summary of experimental results and statistical analysis in CEC2017

Dim	Metrics	L-SHADE	CSO	iL-SHADE	MPEDE	jSO	L-SHADE-cnEpSin	MCSO	OLCSO	TPCSO	DEA ² H ²	SACSO
30	+ / ≈ / −	19/2/8	13/16/0	15/2/12	29/0/0	16/7/6	17/7/5	11/16/2	19/10/0	19/10/0	19/4/6	−
	Avg. ranks:	6.9	5.0	4.2	10.7	6.3	6.6	3.8	6.0	5.1	7.6	3.4
	Corrected ranks:	9	4	3	11	7	8	2	6	5	10	1
50	+ / ≈ / −	19/3/7	8/20/1	17/1/11	28/1/0	19/5/5	20/4/5	10/11/8	25/4/0	19/10/0	19/3/7	−
	Avg. ranks:	6.4	3.8	4.4	10.5	7.0	7.3	3.3	6.8	5.2	7.2	3.7
	Corrected ranks:	6	3	4	11	8	10	1	7	5	9	2

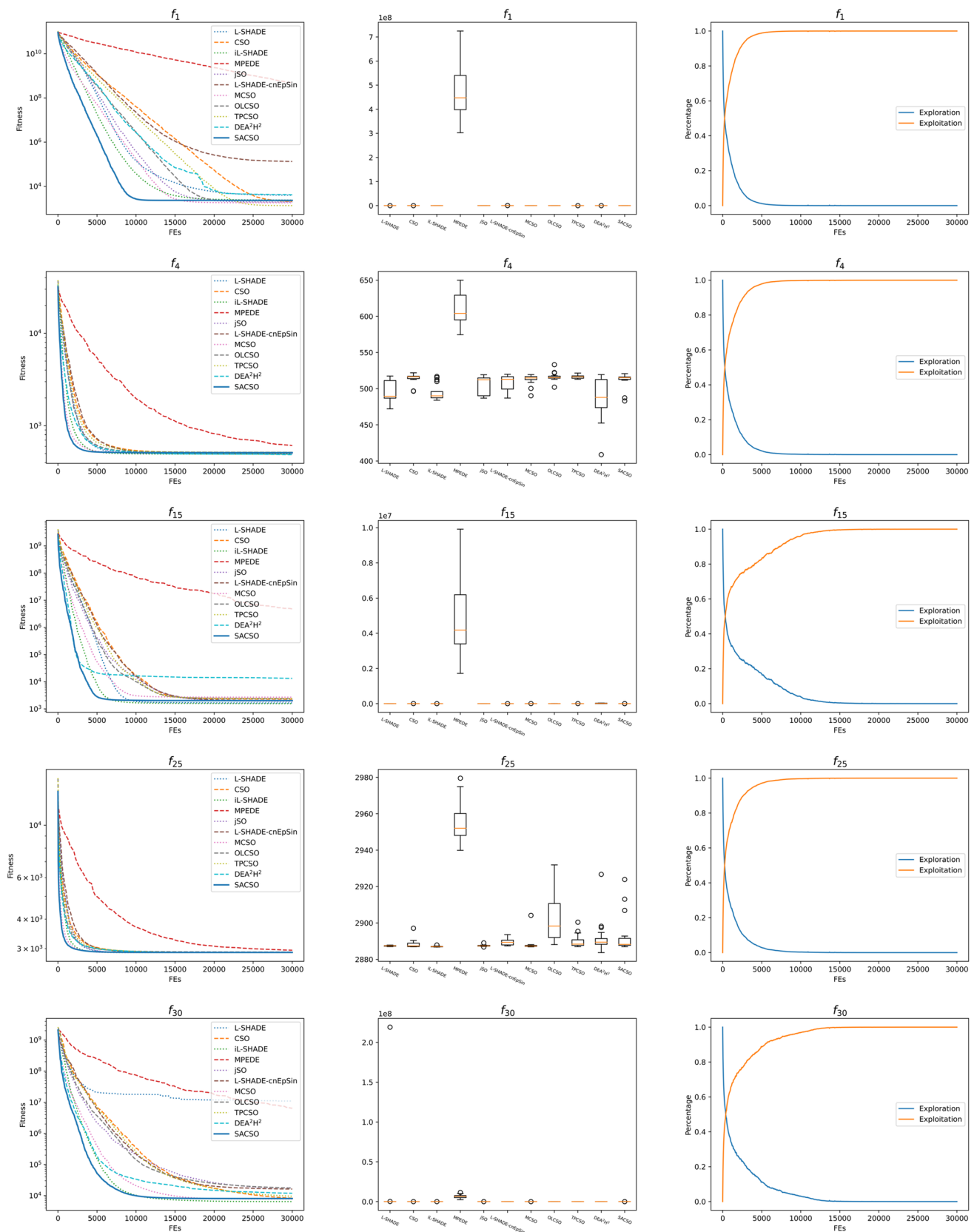


Fig. 6 Convergence curves, boxplots, and the proportion variation between exploration and exploitation in 30-D CEC2017 representative functions

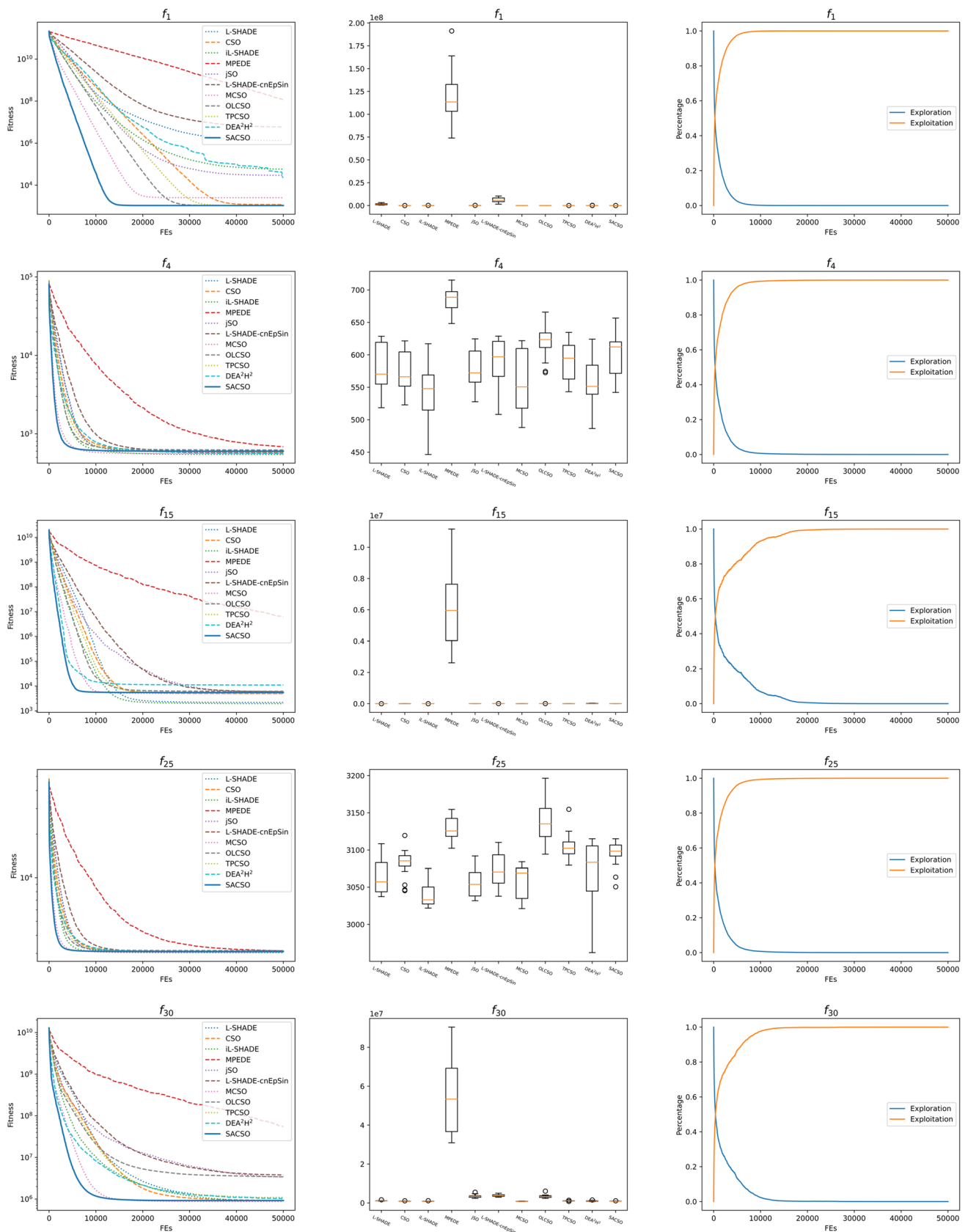


Fig. 7 Convergence curves, boxplots, and the proportion variation between exploration and exploitation in 50-D CEC2017 representative functions

between exploration and exploitation in CEC2022 representative functions.

The competitiveness of the proposed SACSO is also evident in the CEC2022 benchmark, where it achieved average ranks of 3.8 in 10-D and 3.5 in 20-D, which is ranked as the second-best optimizer overall. Considering that L-SHADE-based optimizers are known for being problem-specific and highly effective in CEC competitions - and that iL-SHADE is one of the most prominent variants of L-SHADE - this slight inferiority is acceptable. We still believe SACSO is a highly competitive and superior optimizer for the CEC2022 benchmark.

4.2.3 Experiments on engineering problems

Cantilever beam design problem (CBDP) CBDP aims to optimize the weight of the beam that is fixed at one end and free at the other. A demonstration of CBDP is presented in Fig. 10, and Eq. (6) formulates its mathematical model.

$$\begin{aligned} \min f(X) &= 0.0624(x_1 + x_2 + x_3 + x_4 + x_5) \\ \text{s.t. } g(X) &= \frac{61}{x_1^3} + \frac{37}{x_2^3} + \frac{19}{x_3^3} + \frac{7}{x_4^3} + \frac{1}{x_5^3} - 1 \leq 0 \\ \text{where } 0.01 &\leq x_i \leq 100, i \in \{1, 2, 3, 4, 5\} \end{aligned} \quad (6)$$

The experimental results and statistical analysis of optimizers in CBDP are summarized in Table 5, while ranks, convergence curves, and boxplots of optimizers are presented in Fig. 11. Notice that the best optimizer in the rank bar figure is colored in red, the second-best is in yellow, and the third-best is in green.

The proposed SACSO, DEA^2H^2 , and OLCSSO are ranked as the best, second-best, and third-best optimizers in CBDP. The superior robustness of SACSO can also be observed in the standard deviation summarized in Table 5 and boxplots presented in Fig. 11. Additionally, the high-performance optimizer iL-SHADE confirmed in CEC benchmarks performs poorly in CBDP. This is also practical evidence for the No Free Lunch Theorem (NFLT) in optimization [64].

Corrugated bulkhead design problem (CBHDP) The corrugated bulkhead is a structural component for separating compartments, providing structural support, and resisting bending and buckling. The challenge in CBHDP includes the design in weight, strength, stiffness, and manufacturing costs. Equation (7) formulates the mathematical model of CBHDP.

$$\begin{aligned} \min f(X) &= \frac{5.885x_4(x_1 + x_3)}{x_1 + \sqrt{|x_3^2 - x_2^2|}} \\ \text{s.t. } g_1(X) &= -x_4x_2 \left(0.4x_1 + \frac{x_3}{6}\right) \\ &\quad + 8.94(x_1 + \sqrt{|x_3^2 - x_2^2|}) \leq 0 \end{aligned}$$

Table 4 The summary of experimental results and statistical analysis in CEC2022

Dim	Metrics	L-SHADE	CSO	iL-SHADE	MPEDE	jSO	L-SHADE-cnEpSin	MCSO	OLCSO	TPCSO	DEA^2H^2	SACSO
10	+/-/-;	5/3/4	12/0/0	4/0/8	12/0/0	4/4/4	8/2/2	9/3/0	11/1/0	11/1/0	6/3/3	-
	Avg. ranks:	4.8	8.2	2.7	10.7	4.1	5.4	5.5	6.9	7.0	6.7	3.8
	Corrected ranks:	4	10	1	11	3	5	6	8	9	7	2
20	+/-/-;	10/2/0	9/3/0	5/0/7	12/0/0	6/2/4	7/1/4	6/5/1	10/2/0	8/4/0	8/3/1	-
	Avg. ranks:	8.8	6.9	2.6	10.5	4.5	5.9	4.2	6.4	6.4	6.1	3.5
	Corrected ranks:	10	9	1	11	4	5	3	7	7	6	2

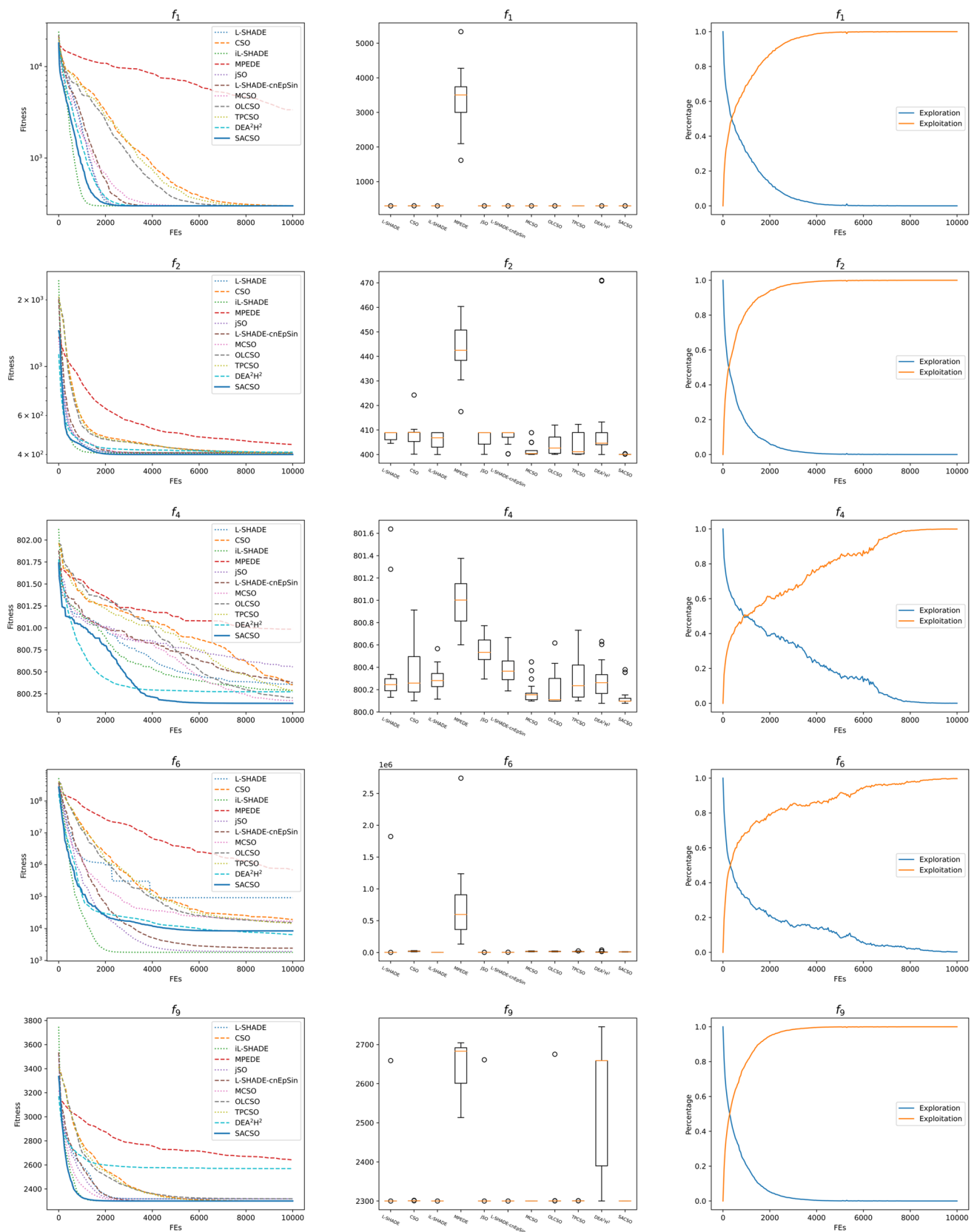


Fig. 8 Convergence curves, boxplots, and the proportion variation between exploration and exploitation in 10-D CEC2022 representative functions

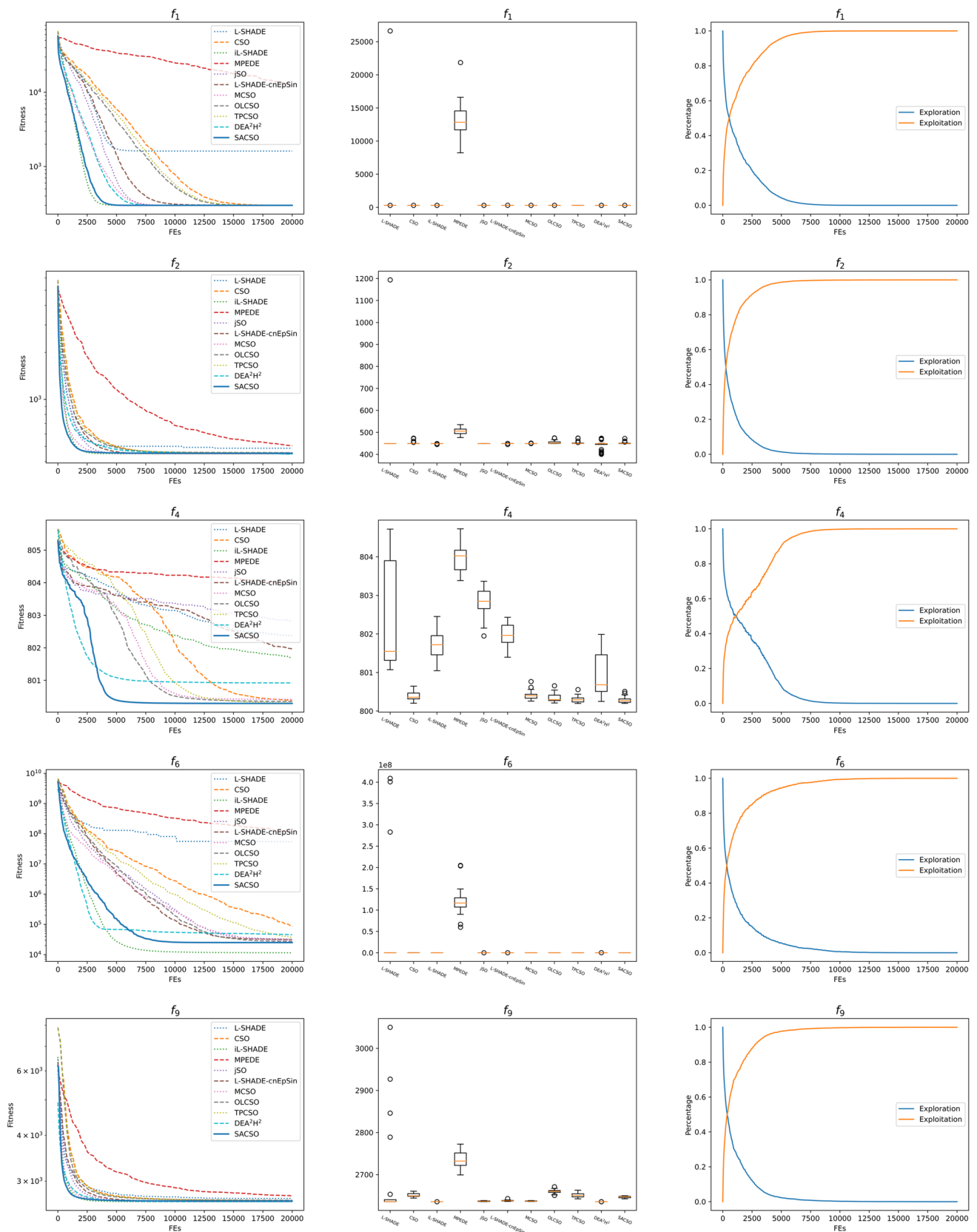
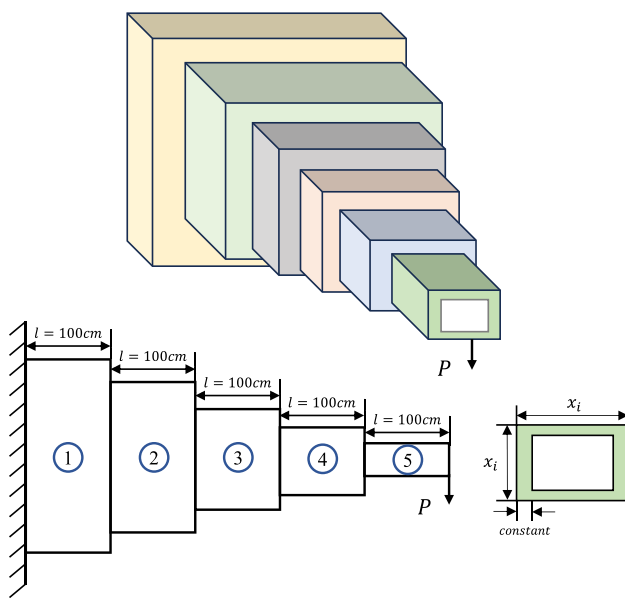


Fig. 9 Convergence curves, boxplots, and the proportion variation between exploration and exploitation in 20-D CEC2022 representative functions

Table 5 Experimental results and statistical analysis in CBHP

MAs	mean	std	best	worst
L-SHADE	1.340410e+00 +	3.377672e−04	1.340057e+00	1.341684e+00
CSO	1.340280e+00 +	1.656135e−04	1.340061e+00	1.340626e+00
iL-SHADE	1.376679e+00 +	1.702132e−02	1.347448e+00	1.405657e+00
MPEDE	1.340360e+00 +	1.431915e−04	1.340137e+00	1.340610e+00
jSO	1.351058e+00 +	6.081428e−03	1.341930e+00	1.363752e+00
L-SHADE-cnEpSin	3.850301e+00 +	4.733259e−01	2.959881e+00	4.500690e+00
MCSO	1.340112e+00 +	1.155282e−04	1.339977e+00	1.340388e+00
OLCSO	1.340049e+00 +	5.028924e−05	1.339983e+00	1.340142e+00
TPCSO	1.340186e+00 +	7.837153e−05	1.340006e+00	1.340382e+00
DEA ² H ²	1.340003e+00 ≈	7.786410e−05	1.339956e+00	1.340251e+00
SACSO	1.339987e+00	1.597581e−05	1.339961e+00	1.340017e+00

**Fig. 10** A demonstration of CBHP

$$\begin{aligned}
 g_2(X) &= -x_4 x_2^2 \left(0.2x_1 + \frac{x_3}{12} \right) \\
 &\quad + 2.2 \left(8.94 \left(x_1 + \sqrt{|x_3^2 - x_2^2|} \right) \right)^{4/3} \leq 0 \\
 g_3(X) &= -x_4 + 0.0156x_1 + 0.15 \leq 0 \\
 g_4(X) &= -x_4 + 0.0156x_3 + 0.15 \leq 0 \\
 g_5(X) &= -x_4 + 1.05 \leq 0 \\
 g_6(X) &= -x_3 + x_2 \leq 0 \\
 \text{where } 0 &\leq x_1, x_2, x_3 \leq 100 \\
 0 &\leq x_4 \leq 5
 \end{aligned} \tag{7}$$

The experimental results and statistical analysis of optimizers in CBHP are summarized in Table 6, while ranks, convergence curves, and boxplots of optimizers are presented in Fig. 12.

The proposed SACSO, MCSO, and OLCSO are ranked as the best, second-best, and third-best optimizers in CBHP, and the experimental results and statistical analysis confirm the remarkable competitiveness and superior scalability of SACSO in various optimization domains. Although the numerical differences between SACSO and other optimizers summarized in Table 6 are tiny, the superior significance always exists. Furthermore, the outstanding convergence speed and accuracy are observable in the convergence curve presented in Fig. 12, which is sufficient to confirm the domination of SACSO in CBHP.

Gear train design problem (GTDP) GTDP focuses on the optimal design of gear systems to achieve the desired speed, torque, and power transmission while minimizing factors including size, weight, noise, and manufacturing costs. A demonstration of GTDP is presented in Fig. 13, and Eq. (8) formulates its mathematical model.

$$\begin{aligned}
 \min f(X) &= \left(\frac{1}{6.931} - \frac{x_3 x_2}{x_1 x_4} \right)^2 \\
 \text{where } x_1, x_2, x_3, x_4 &\in \{12, 13, 14, \dots, 60\}
 \end{aligned} \tag{8}$$

The experimental results and statistical analysis of optimizers in GTDP are summarized in Table 7, while ranks, convergence curves, and boxplots of optimizers are presented in Fig. 14.

DEA²H², the proposed SACSO, and OLCSO are ranked as the top three optimizers for the CBHP, with DEA²H² being the best, followed by SACSO and OLCSO. Notably, DEA²H², which has an average performance among state-of-the-art optimizers in CEC benchmarks, significantly outperforms other optimizers in the GTDP, as evidenced by the convergence curve shown in Fig. 14. Nevertheless, the proposed SACSO is only worse than DEA²H² with statistical significance and significantly better or approximately equal

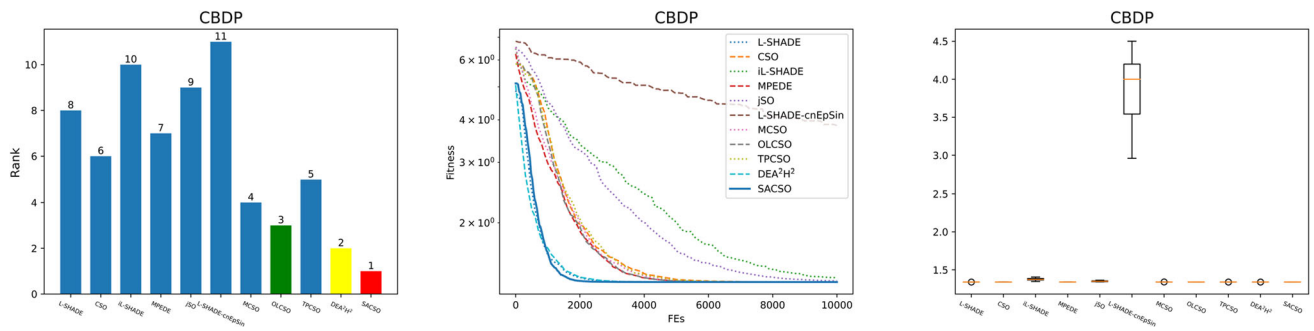


Fig. 11 Ranks, convergence curves, and boxplots of optimizers in CDBP

Table 6 Experimental results and statistical analysis in CBHDP

MAs	mean	std	best	worst
L-SHADE	6.844109e+00 +	5.201663e−04	6.843185e+00	6.845250e+00
CSO	6.844976e+00 +	6.202201e−04	6.843721e+00	6.846010e+00
iL-SHADE	6.859832e+00 +	6.849243e−03	6.850597e+00	6.874232e+00
MPEDE	6.844255e+00 +	5.273243e−04	6.843557e+00	6.845306e+00
jSO	6.852715e+00 +	3.405434e−03	6.847374e+00	6.858576e+00
L-SHADE-cnEpSin	7.345086e+00 +	1.211901e−01	7.108546e+00	7.572175e+00
MCSO	6.843558e+00 +	4.860238e−04	6.843029e+00	6.844979e+00
OLCSO	6.843843e+00 +	1.977456e−04	6.843455e+00	6.844243e+00
TPCSO	6.844394e+00 +	3.645962e−04	6.843796e+00	6.845078e+00
DEA ² H ²	6.851109e+00 +	1.860936e−02	6.842958e+00	6.914245e+00
SACSO	6.842971e+00	2.321927e−05	6.842958e+00	6.843060e+00

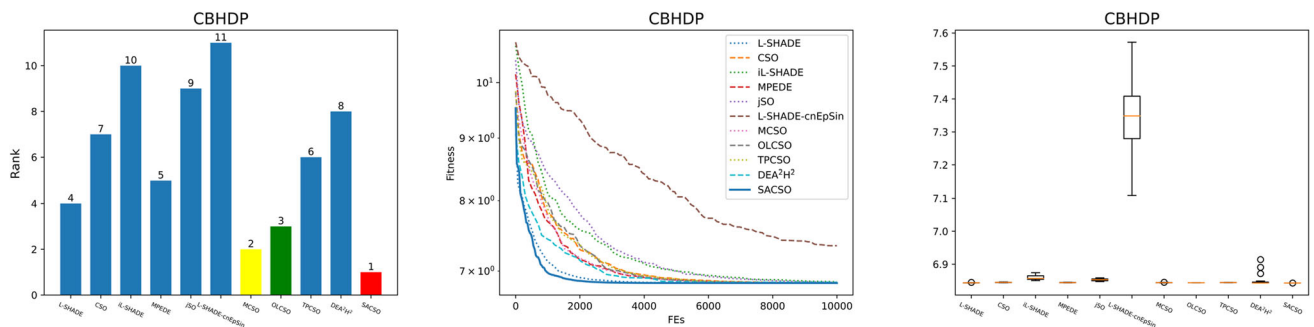


Fig. 12 Ranks, convergence curves, and boxplots of optimizers in CBHDP

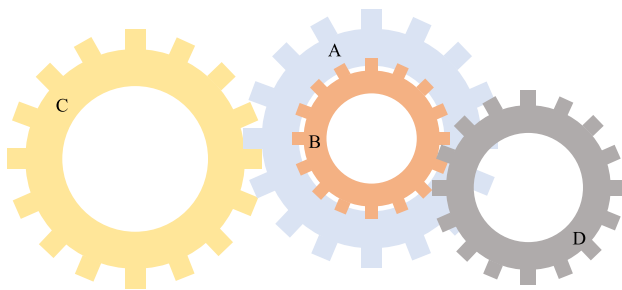


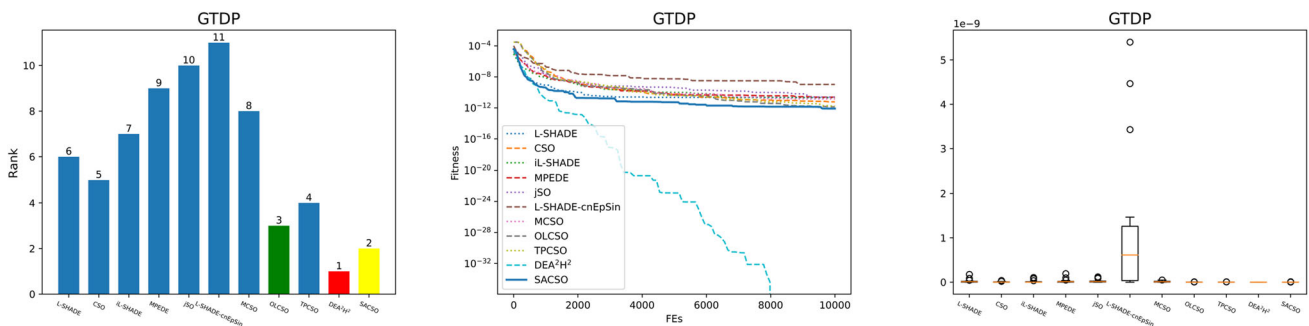
Fig. 13 A demonstration of GTDP

to the rest optimizers. These results demonstrate that SACSO is a successful optimization approach for GTDP.

Piston lever design problem (PLDP) PLDP focuses on designing a piston-lever mechanism to convert linear motion into rotary motion. The objective of PLDP is to optimize the piston and lever dimensions, materials, and configurations to achieve the optimal configuration. A demonstration of PLDP is presented in Fig. 15, and Eq. (9) formulates its mathemat-

Table 7 Experimental results and statistical analysis in GTDP

MAs	mean	std	best	worst
L-SHADE	1.424083e-11 +	1.554302e-11	2.046642e-14	4.959479e-11
CSO	6.086560e-12 +	9.150099e-12	1.390520e-14	3.799726e-11
iL-SHADE	2.018306e-11 ≈	3.700017e-11	3.774823e-32	1.058926e-10
MPEDA	2.450960e-11 +	4.470431e-11	2.145086e-13	1.926030e-10
jSO	2.535331e-11 +	3.520574e-11	6.378717e-15	1.192312e-10
L-SHADE-cnEpSin	1.076690e-09 +	1.519446e-09	6.780499e-13	5.401062e-09
MCSO	2.226194e-11 +	4.011036e-11	4.436578e-14	1.754719e-10
OLCSO	1.395283e-12 ≈	2.126673e-12	1.016883e-14	7.919154e-12
TPCSO	1.774229e-12 ≈	2.704352e-12	1.231610e-14	9.057801e-12
DEA ² H ²	0.000000e+00 −	0.000000e+00	0.000000e+00	0.000000e+00
SACSO	8.087729e-13	2.312753e-12	1.028851e-17	1.067664e-11

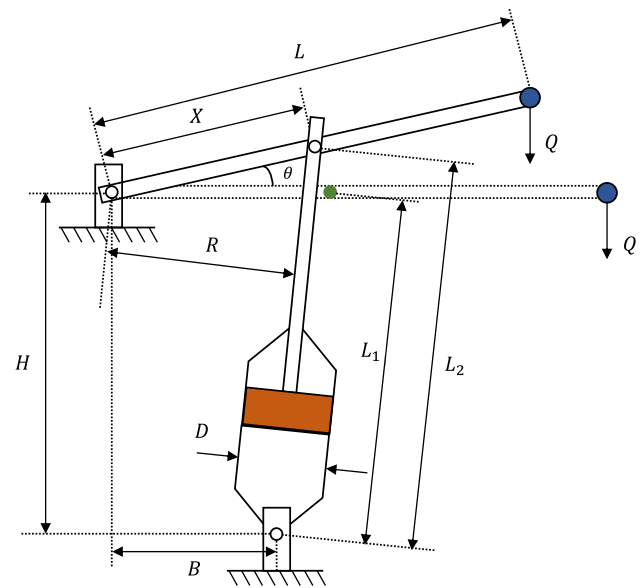
**Fig. 14** Ranks, convergence curves, and boxplots of optimizers in GTDP

ical model. More detailed constraints in PLDP can refer to [65].

$$\begin{aligned}
 \min f(X) &= \frac{1}{4}\pi x_3^2(L_2 - L_1) \\
 \text{s.t. } g_1(X) &= QL\cos(\theta) - R \times F \leq 0 \\
 g_2(X) &= Q(L - x_4) - M_{\max} \leq 0 \\
 g_3(X) &= 1.2(L_2 - L_1) - L_1 \leq 0 \\
 g_4(X) &= \frac{x_3}{2} - x_2 \leq 0 \\
 \text{where } 0.05 &\leq x_1, x_2, x_4 \leq 500 \\
 0.05 &\leq x_3 \leq 120
 \end{aligned}
 \tag{9}$$

The experimental results and statistical analysis of optimizers in PLDP are summarized in Table 8, while ranks, convergence curves, and boxplots of optimizers are presented in Fig. 16.

In PLDP, L-SHADE, MPEDA, and jSO are ranked as the top three optimizers, with SACSO ranked fourth. However, the performance difference between SACSO and the top three optimizers (i.e., L-SHADE, MPEDA, and jSO) is not statistically significant. The robustness of SACSO is fur-

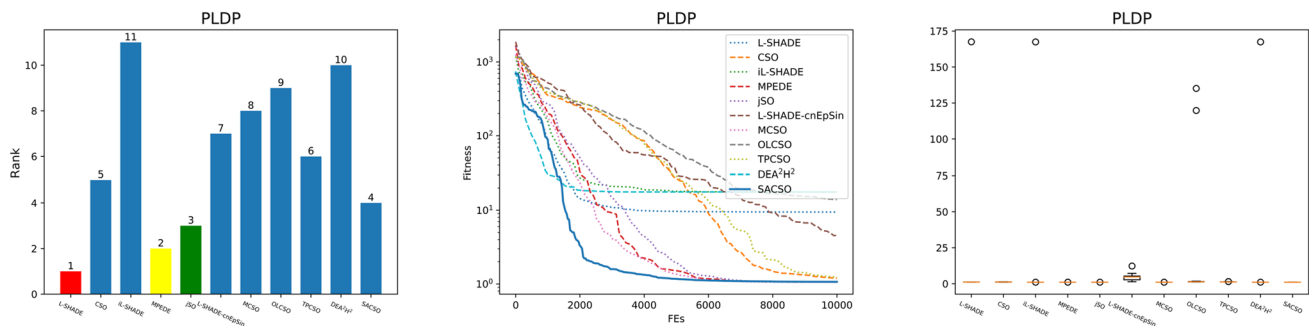
**Fig. 15** A demonstration of PLDP

ther highlighted by the boxplots in Fig. 16, which show no presence of outliers.

Speed reducer design problem (SRDP) SRDP aims at designing a gearbox that reduces the speed of an input shaft

Table 8 Experimental results and statistical analysis in PLDP

MAs	mean	std	best	worst
L-SHADE	1.057899e+00 $\approx$	4.232352e-04	1.057462e+00	1.059219e+00
CSO	1.193975e+00 +	5.960799e-02	1.089468e+00	1.319100e+00
iL-SHADE	1.769933e+01 +	4.992567e+01	1.057404e+00	1.674772e+02
MPEDe	1.058951e+00 $\approx$	3.890630e-03	1.057477e+00	1.074924e+00
jSO	1.059082e+00 $\approx$	1.517480e-03	1.057542e+00	1.062705e+00
L-SHADE-cnEpSin	4.543181e+00 +	2.368582e+00	1.419441e+00	1.225570e+01
MCSO	9.440479e+00 +	3.626485e+01	1.069211e+00	1.675152e+02
OLCSO	1.395326e+01 +	3.792505e+01	1.167831e+00	1.351619e+02
TPCSO	1.238286e+00 +	1.047310e-01	1.114475e+00	1.495081e+00
DEA ² H ²	1.769893e+01 +	4.992460e+01	1.057394e+00	1.674727e+02
SACSO	1.070723e+00	1.203874e-02	1.057649e+00	1.091895e+00

**Fig. 16** Ranks, convergence curves, and boxplots of optimizers in PLDP

while increasing the torque at the output shaft. Equation (10) formulates the mathematical model of SRDP.

$$\begin{aligned}
 \min f(X) &= 0.7854x_1x_2^2(3.3333x_3^2 + 14.9334x_3 - 43.0924) \\
 &\quad - 1.508x_1(x_6^2 + x_7^2) + 7.4777(x_6^3 + x_7^3) + 0.7854(x_4x_6^2 + x_5x_7^2) \\
 \text{s.t. } g_1(X) &= \frac{27}{x_1x_2^2x_3} - 1 \leq 0 \\
 g_2(X) &= \frac{397.5}{x_1x_2^2x_3^2} - 1 \leq 0 \\
 g_3(X) &= \frac{1.93x_4^3}{x_2x_6^4x_3} - 1 \leq 0 \\
 g_4(X) &= \frac{1.93x_5^3}{x_2x_7^4x_3} - 1 \leq 0 \\
 g_5(X) &= \frac{\sqrt{(745x_4/x_2x_3)^2 + 16900000}}{110x_6^3} - 1 \leq 0 \\
 g_6(X) &= \frac{\sqrt{(745x_5/x_2x_3)^2 + 157500000}}{85x_7^3} - 1 \leq 0 \\
 g_7(X) &= \frac{x_2x_3}{40} - 1 \leq 0 \\
 g_8(X) &= \frac{5x_2}{x_1} - 1 \leq 0
 \end{aligned}$$

$$\begin{aligned}
 g_9(X) &= \frac{x_1}{12x_2} - 1 \leq 0 \\
 g_{10}(X) &= \frac{1.5x_6 + 1.9}{x_4} - 1 \leq 0 \\
 g_{11}(X) &= \frac{1.1x_7 + 1.9}{x_5} - 1 \leq 0
 \end{aligned}$$

where $2.6 \leq x_1 \leq 2.6$

$$0.7 \leq x_2 \leq 0.8$$

$$17 \leq x_3 \leq 28$$

$$7.3 \leq x_4x_5 \leq 8.3$$

$$2.9 \leq x_6 \leq 3.9$$

$$5 \leq x_7 \leq 5.5$$

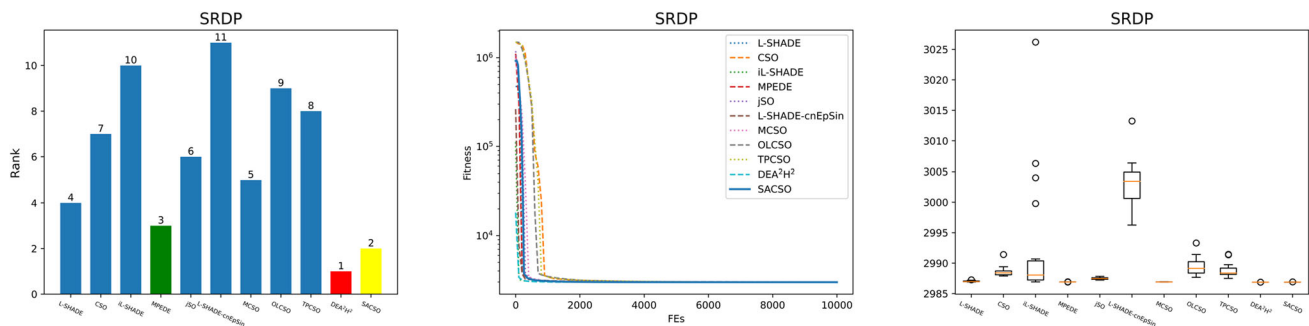
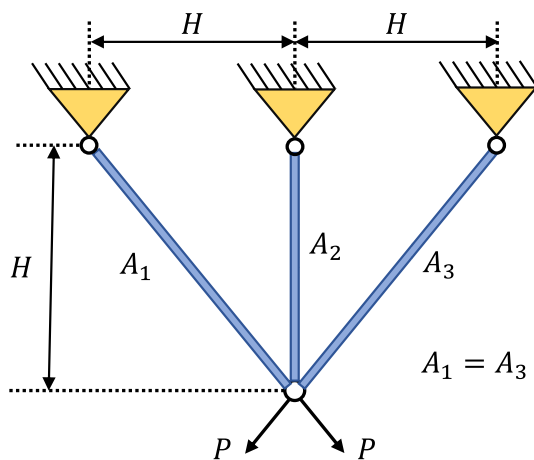
(10)

The experimental results and statistical analysis of optimizers in SRDP are summarized in Table 9, while ranks, convergence curves, and boxplots of optimizers are presented in Fig. 17.

DEA²H², SACSO, and MPEDe are ranked as the best, second-best, and third-best optimizers among the eleven approaches in SRDP, while SACSO is only worse than DEA²H² with statistical significance and significantly outperforms the rest optimization approaches, which demonstrates the competitiveness of SACSO in SRDP.

Table 9 Experimental results and statistical analysis in SRDP

MAs	mean	std	best	worst
L-SHADE	2.986924e+03 +	1.202462e−02	2.986905e+03	2.986946e+03
CSO	2.988594e+03 +	7.622042e−01	2.987898e+03	2.991424e+03
iL-SHADE	2.992216e+03 +	9.626822e+00	2.986936e+03	3.026199e+03
MPEDe	2.986922e+03 +	8.880241e−03	2.986903e+03	2.986942e+03
jSO	2.987484e+03 +	1.639099e−01	2.987222e+03	2.987842e+03
L-SHADE-cnEpSin	3.003067e+03 +	3.681282e+00	2.996264e+03	3.013243e+03
MCSO	2.987018e+03 +	9.823326e−02	2.986910e+03	2.987273e+03
OLCSO	2.989497e+03 +	1.443745e+00	2.987677e+03	2.993307e+03
TPCSO	2.988780e+03 +	1.050078e+00	2.987506e+03	2.991461e+03
DEA ² H ²	2.986885e+03 −	1.116462e−03	2.986884e+03	2.986889e+03
SACSO	2.986896e+03	1.018891e−02	2.986886e+03	2.986928e+03

**Fig. 17** Ranks, convergence curves, and boxplots of optimizers in SRDP**Fig. 18** A demonstration of TBTDP

Three bar truss design problem (TBTDP) The objective of TBTDP is to determine the optimal configuration and dimensions of a simple truss structure while ensuring structural integrity under applied loads. A demonstration of TBTDP is presented in Fig. 18, and Eq. (11) formulates its mathematical

model.

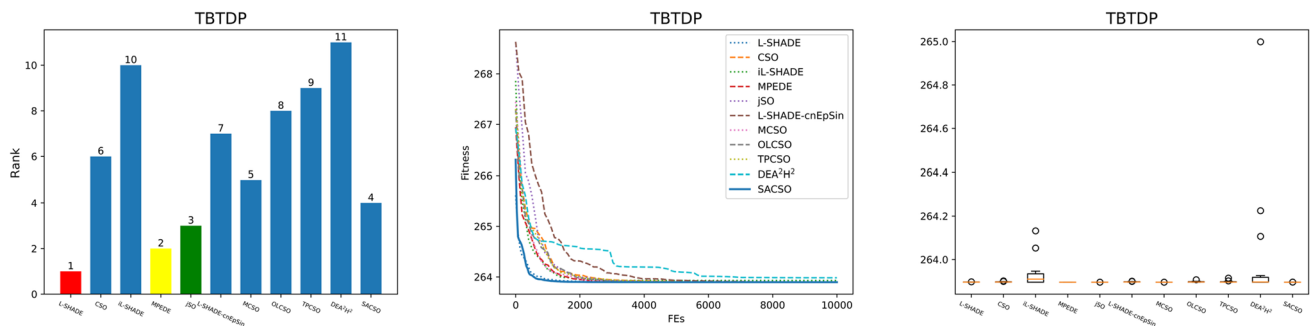
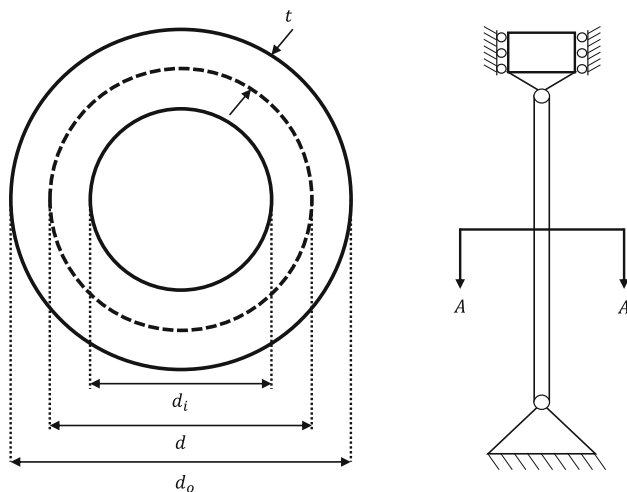
$$\begin{aligned}
 \min f(X) &= (2\sqrt{2}x_1 + x_2) \cdot l \\
 \text{s.t. } g_1(X) &= \frac{\sqrt{2}x_1 + x_2}{\sqrt{2}x_1^2 + 2x_1x_2} P - \sigma \leq 0 \\
 g_2(X) &= \frac{x_2}{\sqrt{2}x_1^2 + 2x_1x_2} P - \sigma \leq 0 \\
 g_3(X) &= \frac{1}{\sqrt{2}x_2 + x_1} P - \sigma \leq 0 \\
 l &= 100 \text{ cm}, P = 2 \text{ kN/cm}^3, \sigma = 2 \text{ kN/cm}^3 \\
 \text{where } 0 &\leq x_1, x_2 \leq 1
 \end{aligned} \tag{11}$$

The experimental results and statistical analysis of optimizers in TBTDP are summarized in Table 10, while ranks, convergence curves, and boxplots of optimizers are presented in Fig. 19.

In TBTDP, L-SHADE, MPEDe, and jSO are ranked as the top three optimization approaches among eleven optimizers, with SACSO ranking fourth. Moreover, L-SHADE, MPEDe, and jSO significantly outperform the proposed SACSO, which indicates that SACSO may not be competitive enough for TBTDP. This observation is also consistent with the NFLT.

Table 10 Experimental results and statistical analysis in TBTD

MAs	mean	std	best	worst
L-SHADE	2.638958e+02 −	6.100996e−07	2.638958e+02	2.638958e+02
CSO	2.638973e+02 +	1.625354e−03	2.638959e+02	2.639022e+02
iL-SHADE	2.639302e+02 +	5.796034e−02	2.638958e+02	2.641321e+02
MPEDA	2.638958e+02 −	3.822805e−07	2.638958e+02	2.638958e+02
jSO	2.638958e+02 −	3.211015e−06	2.638958e+02	2.638959e+02
L-SHADE-cnEpSin	2.638973e+02 +	1.121046e−03	2.638960e+02	2.639006e+02
MCSO	2.638964e+02 +	4.091101e−04	2.638959e+02	2.638976e+02
OLCSO	2.638976e+02 +	2.473672e−03	2.638959e+02	2.639074e+02
TPCSO	2.638979e+02 +	4.067168e−03	2.638959e+02	2.639145e+02
DEA ² H ²	2.639824e+02 +	2.470693e−01	2.638958e+02	2.649986e+02
SACSO	2.638960e+02	1.584644e−04	2.638958e+02	2.638964e+02

**Fig. 19** Ranks, convergence curves, and boxplots of optimizers in TBTD**Fig. 20** A demonstration of TCDP

Tubular column design problem (TCDP) TCDP focuses on designing a tubular column to maximize its load-carrying capacity while minimizing weight, material use, and cost. A demonstration of TCDP is presented in Fig. 20, and Eq. (12) formulates its mathematical model. More detailed constraints

in TCDP can refer to [65].

$$\begin{aligned}
 \min f(X) &= 9.8x_1x_2 + 2x_1 \\
 \text{s.t. } g_1(X) &= \frac{P}{\pi x_1 x_2 \sigma_y} - 1 \leq 0 \\
 g_2(X) &= \frac{8PL^2}{\pi^3 E x_1 x_2 (x_1^2 + x_2^2)} - 1 \leq 0 \\
 \text{where } 2 &\leq x_1 \leq 14 \\
 0.2 &\leq x_2 \leq 0.8
 \end{aligned} \tag{12}$$

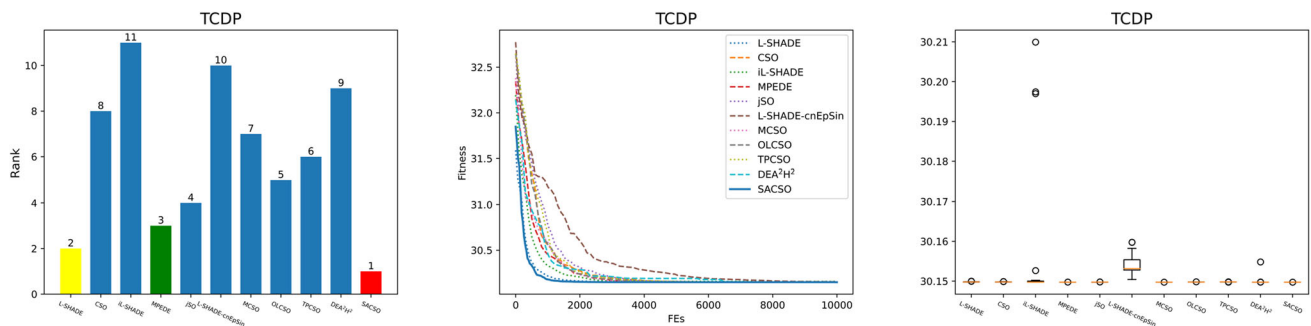
The experimental results and statistical analysis of optimizers in TCDP are summarized in Table 11, while ranks, convergence curves, and boxplots of optimizers are presented in Fig. 21.

The proposed SACSO, L-SHADE, and MPEDA are ranked as the best, second-best, and third-best optimizers in TCDP, while SACSO is significantly better than the other ten optimization approaches, which demonstrates the impressive competitiveness of SACSO.

In summary, the proposed SACSO demonstrates exceptional performance across seven engineering optimization challenges when compared with the other ten state-of-the-art optimizers. As an efficient and effective optimizer, we

Table 11 Experimental results and statistical analysis in TCDP

MAs	mean	std	best	worst
L-SHADE	3.014974e+01 +	1.419579e−06	3.014974e+01	3.014975e+01
CSO	3.014980e+01 +	3.905838e−05	3.014976e+01	3.014991e+01
iL-SHADE	3.015768e+01 +	1.854944e−02	3.014974e+01	3.020994e+01
MPEDE	3.014974e+01 +	3.350930e−06	3.014974e+01	3.014975e+01
jSO	3.014976e+01 +	1.352107e−05	3.014974e+01	3.014980e+01
L-SHADE-cnEpSin	3.015423e+01 +	2.672340e−03	3.015041e+01	3.015977e+01
MCSO	3.014980e+01 +	6.466313e−05	3.014974e+01	3.014999e+01
OLCSO	3.014977e+01 +	2.007254e−05	3.014974e+01	3.014983e+01
TPCSO	3.014979e+01 +	2.162479e−05	3.014975e+01	3.014984e+01
DEA ² H ²	3.015000e+01 +	1.108098e−03	3.014974e+01	3.015483e+01
SACSO	3.014974e+01	2.581317e−07	3.014974e+01	3.014974e+01

**Fig. 21** Ranks, convergence curves, and boxplots of optimizers in TCDP

believe that SACSO has great potential for addressing other real-world engineering applications.

4.2.4 Experiments in HPA design

This section presents the detailed experimental results and statistical analysis of optimizers in HPA design tasks. We conducted the numerical experiments in HPA design instances HPA101, HPA102, HPA103, HPA131, HPA142, and HPA143. Given different dimension sizes and difficulties of HPA tasks are available, dimension sizes $N \in \{4, 5\}$ and difficulties $D \in \{0, 1\}$ are employed as experimental environments. Tables 12 and 13 summarize the detailed experimental results and statistical analysis of optimizers in unconstrained and constrained HPA design instances, respectively. Figures 22 and 23 present the corresponding boxplots to demonstrate the robustness of optimizers.

Based on the experimental results and statistical analysis for unconstrained HPA design tasks summarized in Table 12, SACSO achieves average ranks of 1.8, 1.0, and 1.0, with corrected ranks of 1, 1, and 1 in HPA101, HPA102, and HPA103, respectively, which demonstrates its impressive competitiveness. Additionally, in the statistical analysis summary, SACSO achieved 0/4/0 and 2/2/0 against L-SHADE in HPA101 and HPA103, and 2/2/0 and 2/2/0 against MPEDE

in HPA101 and HPA103, respectively, while being significantly better than the remaining competitor algorithms in other instances.

Similarly, the dominant performance of SACSO is also evident in constrained HPA design tasks, as summarized in Table 13, where the average ranks of SACSO in HPA131, HPA142, and HPA143 are 1.5, 1.0, and 1.0, with corrected ranks of 1, 1, and 1. Furthermore, SACSO achieved 2/2/0 against both L-SHADE and MPEDE in HPA131. These results confirm that the proposed SACSO is a highly suitable and high-performance optimizer for HPA design. We believe that SACSO has great potential for tackling more challenging HPA design tasks, and its variants can be extended to multi-objective HPA design problems.

4.2.5 Ablation experiments in CEC benchmarks

To investigate the performance of introduced strategies independently, we conduct ablation experiments in CEC benchmarks. Here, the representation of abbreviations are as follows: CSO: the original CSO algorithm; SCSO: CSO + parameter sorting strategy; LCSO: CSO + linear population reduction scheme; SACSO: CSO + integrated two strategies. CSO is employed as the basic optimizer in the statistical analysis implemented by the Holm multiple comparison test.

Table 12 Experimental results and statistical analysis of optimizers in unconstrained HPA design tasks

Tasks		L-SHADE	CSO	iL-SHADE	MPEDA	jSO	L-SHADE-cnEpSin	MCSO	OLCSO	TPCSO	DEA ² H ²	SACSO
HPA101	N=4, D=0	mean	2.256e+01 +	2.296e+01 +	2.253e+01 ≈	2.285e+01 +	2.411e+01 +	2.311e+01 +	2.312e+01 +	2.311e+01 +	2.319e+01 +	2.254e+01
		std	8.394e-02	2.663e-01	5.830e-02	1.902e-01	4.222e-01	1.593e-01	1.992e-01	1.593e-01	4.384e-01	9.384e-02
	N=4, D=1	mean	2.238e+01 ≈	2.326e+01 +	2.240e+01 +	2.322e+01 +	2.508e+01 +	2.314e+01 +	2.305e+01 +	2.314e+01 +	2.334e+01 +	2.228e+01
		std	1.329e-01	3.570e-01	1.465e-01	3.340e-01	3.640e-01	2.269e-01	2.397e-01	2.269e-01	2.993e-01	2.036e-01
	N=5, D=0	mean	2.269e+01 ≈	2.318e+01 +	2.274e+01 ≈	2.294e+01 +	2.475e+01 +	2.359e+01 +	2.358e+01 +	2.359e+01 +	2.356e+01 +	2.279e+01
		std	1.180e-01	2.759e-01	1.848e-01	1.541e-01	3.983e-01	2.522e-01	2.851e-01	2.522e-01	3.800e-01	2.056e-01
N=5, D=1	mean	2.267e+01 ≈	2.392e+01 +	2.396e+01 +	2.277e+01 +	2.365e+01 +	2.587e+01 +	2.390e+01 +	2.375e+01 +	2.390e+01 +	2.398e+01 +	2.256e+01
	std	1.794e-01	2.823e-01	5.222e-01	2.573e-01	2.509e-01	5.501e-01	2.927e-01	3.751e-01	2.927e-01	6.097e-01	1.964e-01
	+ / ≈ / -	0/4/0	4/0/0	4/0/0	2/2/0	4/0/0	4/0/0	4/0/0	4/0/0	4/0/0	4/0/0	-
	Avg. ranks	2.0	7.8	7.0	2.3	5.0	11.0	6.5	6.3	7.5	9.0	1.8
	Corrected ranks	2	9	7	3	4	11	6	5	8	10	1
HPA102	N=4, D=0	mean	2.170e+02 +	2.207e+02 +	2.169e+02 +	2.200e+02 +	2.315e+02 +	2.204e+02 +	2.198e+02 +	2.204e+02 +	2.213e+02 +	2.161e+02
		std	5.981e-01	1.646e+00	4.432e-01	1.240e+00	3.080e+00	1.777e+00	1.340e+00	1.777e+00	4.186e+00	3.973e-01
	N=4, D=1	mean	2.182e+02 +	2.258e+02 +	2.175e+02 +	2.256e+02 +	2.422e+02 +	2.254e+02 +	2.240e+02 +	2.254e+02 +	2.286e+02 +	2.156e+02
		std	1.771e+00	3.507e+00	1.624e+00	2.912e+00	6.231e+00	4.069e+00	2.346e+00	4.069e+00	4.522e+00	1.746e+00
	N=5, D=0	mean	2.160e+02 +	2.195e+02 +	2.162e+02 +	2.200e+02 +	2.324e+02 +	2.195e+02 +	2.192e+02 +	2.195e+02 +	2.206e+02 +	2.151e+02
		std	6.935e-01	1.245e+00	6.224e-01	1.876e+00	2.286e+00	1.659e+00	1.120e+00	1.659e+00	3.688e+00	4.620e-01
N=5, D=1	mean	2.193e+02 +	2.278e+02 +	2.300e+02 +	2.188e+02 +	2.263e+02 +	2.452e+02 +	2.262e+02 +	2.258e+02 +	2.262e+02 +	2.300e+02 +	2.171e+02
	std	1.953e+00	2.771e+00	5.105e+00	1.622e+00	2.507e+00	5.133e+00	2.924e+00	2.407e+00	2.924e+00	4.235e+00	2.738e+00
	+ / ≈ / -	4/0/0	4/0/0	4/0/0	4/0/0	4/0/0	4/0/0	4/0/0	4/0/0	4/0/0	4/0/0	-
	Avg. ranks	2.8	7.8	10.0	2.3	6.8	11.0	5.3	4.0	6.3	9.0	1.0
	Corrected ranks	3	8	10	2	7	11	5	4	6	9	1
HPA103	N=4, D=0	mean	-1.193e+01 ≈	-1.167e+01 +	-1.189e+01 ≈	-1.164e+01 +	-1.070e+01 +	-1.141e+01 +	-1.137e+01 +	-1.141e+01 +	-1.144e+01 +	-1.199e+01
		std	1.245e-01	1.998e-01	1.721e-01	1.337e-01	4.142e-01	1.630e-01	1.909e-01	1.630e-01	2.454e-01	1.039e-01
	N=4, D=1	mean	-1.161e+01 +	-1.142e+01 +	-1.163e+01 +	-1.131e+01 +	-1.053e+01 +	-1.096e+01 +	-1.104e+01 +	-1.096e+01 +	-1.115e+01 +	-1.178e+01
		std	1.643e-01	2.993e-01	1.319e-01	2.503e-01	2.657e-01	2.820e-01	2.754e-01	2.820e-01	3.501e-01	1.329e-01
	N=5, D=0	mean	-1.187e+01 ≈	-1.163e+01 +	-1.191e+01 ≈	-1.152e+01 +	-1.061e+01 +	-1.125e+01 +	-1.123e+01 +	-1.125e+01 +	-1.153e+01 +	-1.192e+01
		std	1.338e-01	2.757e-01	1.525e-01	1.902e-01	2.882e-01	2.025e-01	2.669e-01	2.025e-01	3.951e-01	1.237e-01
N=5, D=1	mean	-1.156e+01 +	-1.090e+01 +	-1.126e+01 +	-1.157e+01 +	-1.117e+01 +	-1.049e+01 +	-1.093e+01 +	-1.094e+01 +	-1.093e+01 +	-1.083e+01 +	-1.177e+01
	std	1.252e-01	1.723e-01	3.728e-01	1.385e-01	1.993e-01	3.043e-01	2.066e-01	2.047e-01	2.066e-01	5.272e-01	1.362e-01
	+ / ≈ / -	2/2/0	4/0/0	4/0/0	2/2/0	4/0/0	4/0/0	4/0/0	4/0/0	4/0/0	4/0/0	-
	Avg. ranks	2.8	8.5	4.0	2.3	5.3	11.0	7.8	8.0	8.8	6.8	1.0
	Corrected ranks	3	9	4	2	5	11	7	8	10	6	1

Table 13 Experimental results and statistical analysis of optimizers in constrained HPA design tasks

Tasks		L-SHADE	CSO	iL-SHADE	MPEDA	jSO	L-SHADE-cnEpSin	MCSO	OLCSO	TPCSO	DEA ² H ²	SACSO
HPA131	N=4, D=0	mean	2.268e+01	2.319e+01	2.268e+01	2.342e+01	2.469e+01	2.341e+01	2.331e+01	2.341e+01	2.346e+01	2.263e+01
		std	1.237e-01	3.258e-01	1.473e-01	2.832e-01	5.993e-01	3.596e-01	3.599e-01	3.596e-01	2.899e-01	1.396e-01
	N=4, D=1	mean	2.267e+01	2.378e+01	2.265e+01	2.376e+01	2.569e+01	2.343e+01	2.328e+01	2.343e+01	2.387e+01	2.235e+01
		std	2.607e-01	6.729e-01	2.198e-01	4.047e-01	4.496e-01	2.074e-01	2.023e-01	2.074e-01	8.675e-01	2.374e-01
N=5, D=0	mean	2.294e+01	2.406e+01	2.348e+01	2.289e+01	2.348e+01	2.532e+01	2.399e+01	2.395e+01	2.399e+01	2.378e+01	2.299e+01
		std	2.233e-01	5.234e-01	1.903e-01	2.882e-01	4.415e-01	3.473e-01	2.450e-01	3.473e-01	6.031e-01	2.329e-01
	N=5, D=1	mean	2.310e+01	2.434e+01	2.302e+01	2.394e+01	2.631e+01	2.442e+01	2.432e+01	2.442e+01	2.416e+01	2.280e+01
		std	2.645e-01	7.037e-01	2.842e-01	3.649e-01	6.074e-01	4.115e-01	3.725e-01	4.115e-01	5.970e-01	2.384e-01
+/≈/-			2/2/0	4/0/0	2/2/0	4/0/0	4/0/0	4/0/0	4/0/0	4/0/0	4/0/0	-
Avg. ranks		2.5	7.8	6.5	2.0	5.25	11.0	7.5	5.8	8.5	7.8	1.5
Corrected ranks		3	8	6	2	4	11	7	5	10	9	1
HPA142	N=4, D=0	mean	2.215e+02	2.271e+02	2.208e+02	2.294e+02	2.456e+02	2.245e+02	2.239e+02	2.245e+02	2.262e+02	2.179e+02
		std	2.216e+00	6.267e+00	1.599e+00	3.416e+00	6.819e+00	1.793e+00	2.102e+00	1.793e+00	4.783e+00	9.719e-01
	N=4, D=1	mean	2.267e+02	2.444e+02	2.241e+02	2.395e+02	2.648e+02	2.329e+02	2.324e+02	2.329e+02	2.385e+02	2.205e+02
		std	3.160e+00	1.253e+01	3.173e+00	4.238e+00	7.617e+00	4.252e+00	4.077e+00	4.252e+00	9.815e+00	3.611e+00
N=5, D=0	mean	2.204e+02	2.258e+02	2.282e+02	2.193e+02	2.287e+02	2.453e+02	2.253e+02	2.246e+02	2.253e+02	2.236e+02	2.165e+02
		std	2.637e+00	4.178e+00	1.377e+00	3.283e+00	7.375e+00	2.374e+00	2.548e+00	2.374e+00	5.709e+00	8.132e-01
	N=5, D=1	mean	2.263e+02	2.427e+02	2.277e+02	2.418e+02	2.664e+02	2.343e+02	2.333e+02	2.343e+02	2.407e+02	2.219e+02
		std	3.060e+00	8.275e+00	3.858e+00	5.149e+00	5.237e+00	2.564e+00	3.812e+00	2.564e+00	4.429e+00	4.298e+00
+/≈/-			4/0/0	4/0/0	4/0/0	4/0/0	4/0/0	4/0/0	4/0/0	4/0/0	4/0/0	-
Avg. ranks		2.3	8.0	4.5	2.8	5.0	11.0	7.8	8.5	8.8	6.5	1.0
Corrected ranks		2	8	4	3	5	11	7	9	10	6	1
HPA143	N=4, D=0	mean	-1.177e+01	-1.150e+01	-1.181e+01	-1.135e+01	-1.039e+01	-1.127e+01	-1.128e+01	-1.127e+01	-1.136e+01	-1.187e+01
		std	1.314e-01	1.959e-01	1.142e-01	2.016e-01	3.510e-01	1.919e-01	1.868e-01	1.919e-01	2.366e-01	1.558e-01
	N=4, D=1	mean	-1.149e+01	-1.116e+01	-1.148e+01	-1.110e+01	-1.027e+01	-1.078e+01	-1.076e+01	-1.078e+01	-1.075e+01	-1.173e+01
		std	1.459e-01	3.441e-01	1.622e-01	2.561e-01	2.558e-01	2.282e-01	2.421e-01	2.282e-01	6.785e-01	2.220e-01
N=5, D=0	mean	-1.175e+01	-1.105e+01	-1.136e+01	-1.174e+01	-1.134e+01	-1.050e+01	-1.110e+01	-1.104e+01	-1.110e+01	-1.113e+01	-1.185e+01
		std	9.751e-02	3.902e-01	1.148e-01	1.946e-01	2.605e-01	1.398e-01	2.203e-01	1.398e-01	3.772e-01	1.137e-01
	N=5, D=1	mean	-1.143e+01	-1.088e+01	-1.142e+01	-1.091e+01	-1.019e+01	-1.073e+01	-1.074e+01	-1.073e+01	-1.089e+01	-1.160e+01
		std	1.810e-01	4.949e-01	2.013e-01	1.854e-01	2.309e-01	1.800e-01	1.604e-01	1.800e-01	2.030e-01	2.203e-01
+/≈/-			4/0/0	4/0/0	4/0/0	4/0/0	4/0/0	4/0/0	4/0/0	4/0/0	4/0/0	-
Avg. ranks		2.3	8.0	4.5	2.8	5.0	11.0	7.8	8.5	8.8	6.5	1.0
Corrected ranks		2	8	4	3	5	11	7	9	10	6	1

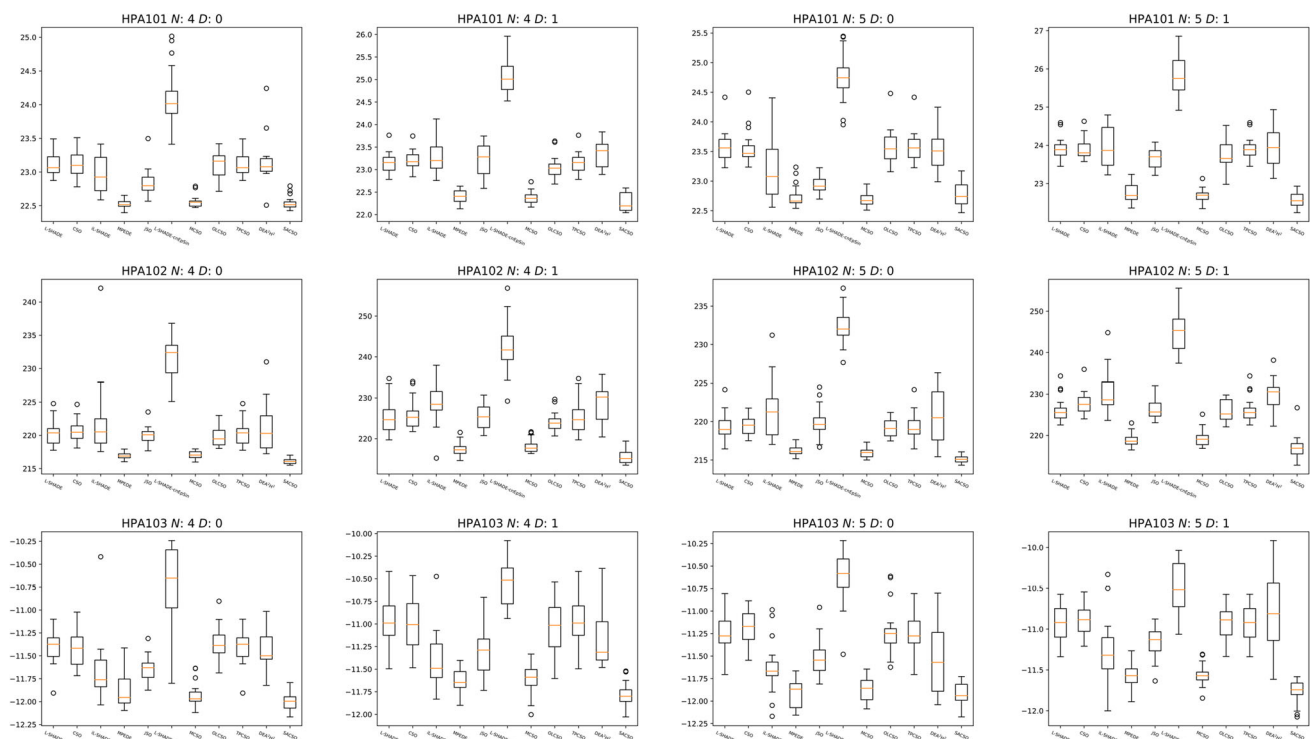


Fig. 22 Boxplots of optimizers in unconstrained HPA design tasks

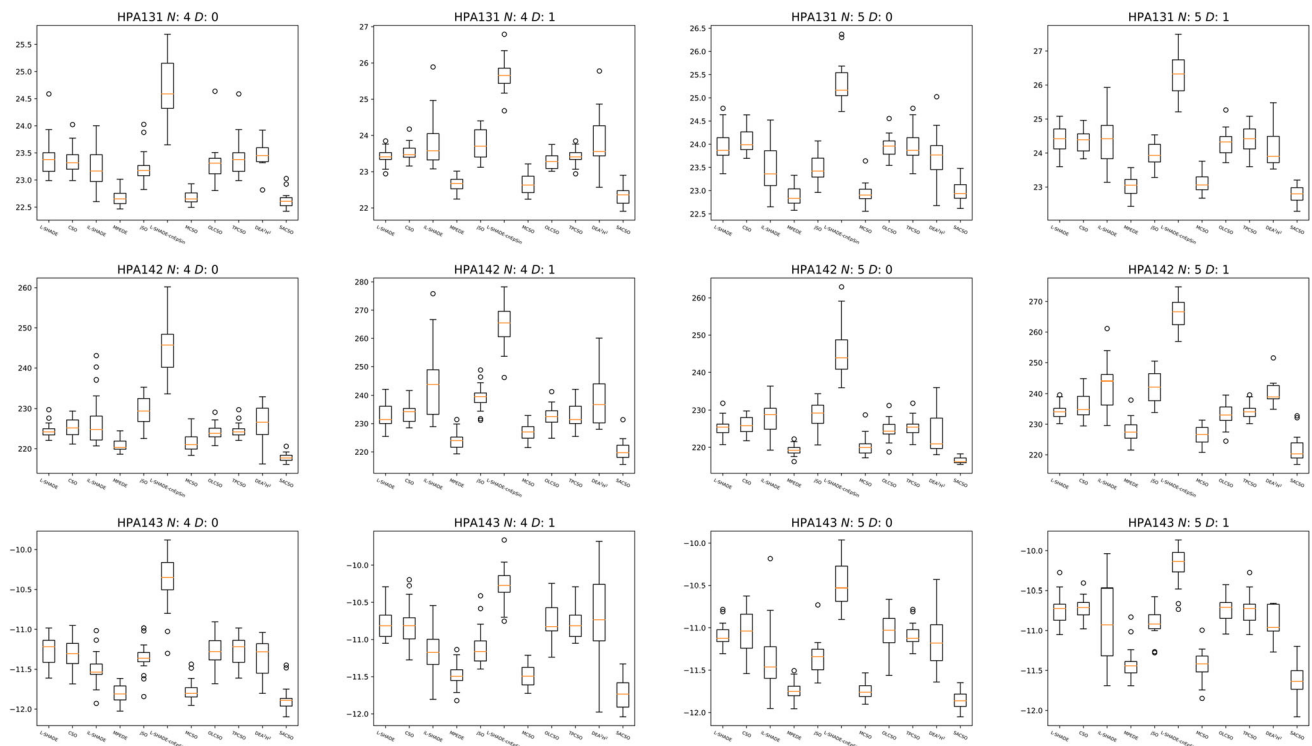


Fig. 23 Boxplots of optimizers in constrained HPA design tasks

Table 14 Results of ablation experiments in 30-D CEC2017

Func	CSO		SCSO		LCSO		SACSO	
	mean	std	mean	std	mean	std	mean	std
f_1	2.075e+03	1.918e+03	2.123e+03 $\approx$	1.903e+03	1.942e+03 $\approx$	1.848e+03	2.321e+03 $\approx$	1.606e+03
f_3	4.612e+04	1.006e+04	3.876e+04 $-$	9.196e+03	4.493e+04 $\approx$	9.026e+03	2.922e+04 $-$	5.283e+03
f_4	5.142e+02	6.198e+00	5.157e+02 $\approx$	4.633e+00	5.158e+02 $\approx$	2.424e+00	5.122e+02 $\approx$	9.278e+00
f_5	5.173e+02	1.234e+01	5.225e+02 $\approx$	1.008e+01	5.133e+02 $\approx$	2.730e+00	5.124e+02 $-$	2.162e+00
f_6	6.000e+02	2.024e-03	6.000e+02 $-$	8.236e-04	6.000e+02 $-$	2.655e-05	6.000e+02 $-$	1.319e-04
f_7	8.645e+02	2.162e+01	7.968e+02 $-$	1.004e+01	7.374e+02 $-$	1.657e+00	7.384e+02 $-$	2.223e+00
f_8	8.168e+02	1.114e+01	8.131e+02 $\approx$	3.828e+01	8.128e+02 $\approx$	2.837e+00	8.123e+02 $\approx$	2.637e+00
f_9	9.000e+02	9.897e-05	9.000e+02 $+$	1.788e-05	9.000e+02 $-$	9.902e-06	9.000e+02 $-$	1.275e-07
f_{10}	5.084e+03	6.682e+02	5.251e+03 $\approx$	7.043e+02	2.702e+03 $-$	4.428e+02	2.349e+03 $-$	4.143e+02
f_{11}	1.205e+03	3.480e+01	1.205e+03 $\approx$	3.033e+01	1.188e+03 $\approx$	4.739e+01	1.176e+03 $\approx$	3.403e+01
f_{12}	8.192e+05	5.249e+05	7.304e+05 $\approx$	4.610e+05	8.445e+05 $\approx$	3.068e+05	7.841e+05 $\approx$	1.896e+05
f_{13}	1.265e+04	4.571e+03	1.165e+04 $\approx$	3.747e+03	1.264e+04 $\approx$	5.191e+03	1.147e+04 $\approx$	2.592e+03
f_{14}	1.502e+05	7.358e+04	8.286e+04 $-$	6.467e+04	1.553e+05 $\approx$	1.602e+05	1.107e+05 $-$	6.670e+04
f_{15}	2.281e+03	1.107e+03	2.164e+03 $\approx$	8.306e+02	2.391e+03 $\approx$	9.703e+02	2.022e+03 $-$	4.738e+02
f_{16}	1.913e+03	2.365e+02	1.749e+03 $-$	1.815e+02	2.055e+03 $\approx$	1.764e+02	1.862e+03 $\approx$	1.926e+02
f_{17}	1.820e+03	9.878e+01	1.810e+03 $\approx$	9.598e+01	1.812e+03 $\approx$	8.810e+01	1.808e+03 $-$	1.051e+02
f_{18}	3.566e+05	1.825e+05	3.784e+05 $\approx$	3.444e+05	3.060e+05 $\approx$	2.213e+05	2.796e+05 $\approx$	1.495e+05
f_{19}	5.600e+03	2.231e+03	5.806e+03 $\approx$	2.598e+03	5.428e+03 $\approx$	2.527e+03	5.303e+03 $\approx$	1.145e+03
f_{20}	2.154e+03	3.956e+00	2.135e+03 $-$	8.099e+01	2.206e+03 $+$	9.885e+01	2.154e+03 $\approx$	6.328e+00
f_{21}	2.314e+03	5.098e+00	2.312e+03 $\approx$	4.473e+00	2.312e+03 $-$	3.588e+00	2.310e+03 $-$	3.459e+00
f_{22}	2.300e+03	3.219e-03	2.300e+03 $-$	1.838e-03	2.300e+03 $-$	2.415e-05	2.300e+03 $-$	9.327e-04
f_{23}	2.663e+03	6.286e+00	2.657e+03 $-$	5.807e+00	2.662e+03 $\approx$	7.230e+00	2.662e+03 $\approx$	6.943e+00
f_{24}	2.828e+03	3.075e+00	2.828e+03 $\approx$	3.157e+00	2.834e+03 $+$	7.000e+00	2.824e+03 $-$	3.553e+00
f_{25}	2.888e+03	2.283e+00	2.888e+03 $\approx$	6.170e-01	2.889e+03 $\approx$	2.536e+00	2.893e+03 $\approx$	9.814e+00
f_{26}	3.693e+03	2.209e+02	3.552e+03 $\approx$	6.640e+01	3.657e+03 $\approx$	2.128e+02	3.522e+03 $\approx$	4.227e+02
f_{27}	3.219e+03	6.819e+00	3.212e+03 $\approx$	6.911e+00	3.228e+03 $\approx$	9.230e+00	3.220e+03 $\approx$	7.162e+00
f_{28}	3.234e+03	1.283e+01	3.227e+03 $\approx$	1.296e+01	3.252e+03 $\approx$	1.321e+01	3.239e+03 $\approx$	1.734e+01
f_{29}	3.460e+03	1.029e+02	3.415e+03 $-$	6.300e+01	3.430e+03 $\approx$	1.079e+02	3.407e+03 $-$	7.813e+01
f_{30}	8.290e+03	1.137e+03	7.324e+03 $-$	6.178e+02	8.506e+03 $\approx$	1.320e+03	8.075e+03 $\approx$	1.146e+03
+/ $\approx$ / $-$	$-$		1/18/10		2/21/6		0/16/13	
Avg. ranks	3.2		2.3		2.8		1.7	
Corrected ranks	4		2		3		1	

Tables 14, 15, 16, and 17 summarize the results of ablation experiments in CEC2017 and CEC2022.

From the ablation experiments in CEC2017 and CEC2022, both the parameter sorting strategy and linear population reduction scheme can accelerate the optimization convergence and improve the optimization accuracy for the original CSO. The experimental results and statistical analysis summarized in Tables 14, 15, 16, and 17 confirm that the parameter sorting strategy contributes more improvement to SACSO in CEC2017, while the integrated two strategies contribute approximately equally in CEC2022. Consequently, we suggest combining both strategies in SACSO simultaneously to handle optimization challenges.

5 Conclusion

This paper presents a Self-Adaptive Competitive Swarm Optimizer (SACSO) to address complex optimization challenges. We focus on two critical parameters in CSO: population size N and scaling factor φ . To automatically and adaptively adjust these parameters during optimization, we integrate a parameter sorting scheme and a linear population reduction strategy into CSO. The parameter sorting scheme assigns smaller scaling factors to well-performing particles to promote the exploitative search, while larger scaling factors are allocated to poorly performing particles to enhance explorative capabilities. Additionally, the

Table 15 Results of ablation experiments in 50-D CEC2017

Func	CSO		SCSO		LCSO		SACSO	
	mean	std	mean	std	mean	std	mean	std
f_1	1.205e+03	1.411e+03	1.627e+03 $\approx$	1.881e+03	1.308e+03 $\approx$	1.672e+03	1.087e+03 $\approx$	7.179e+02
f_3	1.028e+05	1.279e+04	8.777e+04 $-$	1.575e+04	9.879e+04 $\approx$	1.422e+04	7.058e+04 $-$	1.240e+04
f_4	5.736e+02	3.137e+01	5.583e+02 $\approx$	3.694e+01	5.750e+02 $\approx$	2.818e+01	5.996e+02 $\approx$	3.065e+01
f_5	5.237e+02	4.451e+00	5.255e+02 $\approx$	5.550e+00	5.252e+02 $\approx$	5.256e+00	5.266e+02 $\approx$	2.933e+00
f_6	6.000e+02	1.782e-04	6.000e+02 $\approx$	2.566e-04	6.000e+02 $-$	1.007e-05	6.000e+02 $-$	8.827e-05
f_7	8.217e+02	4.548e+01	1.072e+03 $+$	1.656e+01	7.634e+02 $-$	2.944e+00	7.649e+02 $-$	2.495e+00
f_8	8.229e+02	2.952e+00	8.250e+02 $\approx$	2.589e+00	8.289e+02 $+$	4.505e+00	8.255e+02 $\approx$	4.212e+00
f_9	9.003e+02	2.741e-01	9.001e+02 $-$	1.615e-01	9.006e+02 $\approx$	7.146e-01	9.003e+02 $\approx$	3.755e-01
f_{10}	3.208e+03	6.819e+02	3.232e+03 $\approx$	7.630e+02	3.319e+03 $\approx$	7.010e+02	3.110e+03 $\approx$	4.089e+02
f_{11}	1.919e+03	3.839e+02	1.820e+03 $\approx$	4.090e+02	1.739e+03 $-$	3.249e+02	1.783e+03 $-$	2.706e+02
f_{12}	2.367e+06	9.006e+05	1.805e+06 $\approx$	5.670e+05	2.587e+06 $\approx$	9.887e+05	2.206e+06 $\approx$	5.997e+05
f_{13}	2.458e+03	1.059e+03	2.003e+03 $\approx$	6.360e+02	2.493e+03 $\approx$	1.014e+03	2.420e+03 $\approx$	6.385e+02
f_{14}	3.197e+05	1.738e+05	1.431e+05 $-$	7.485e+04	3.444e+05 $\approx$	1.849e+05	3.666e+05 $\approx$	2.205e+05
f_{15}	4.980e+03	2.195e+03	4.865e+03 $\approx$	2.521e+03	4.349e+03 $\approx$	2.400e+03	5.498e+03 $\approx$	1.606e+03
f_{16}	2.194e+03	1.697e+02	2.052e+03 $-$	1.588e+02	2.189e+03 $\approx$	2.089e+02	2.138e+03 $\approx$	1.413e+02
f_{17}	2.061e+03	1.802e+02	2.019e+03 $\approx$	1.314e+02	2.182e+03 $\approx$	2.125e+02	2.147e+03 $\approx$	1.940e+02
f_{18}	2.137e+06	6.512e+05	1.875e+06 $\approx$	7.599e+05	2.135e+06 $\approx$	5.841e+05	1.762e+06 $-$	4.040e+05
f_{19}	1.567e+04	3.467e+03	1.623e+04 $\approx$	3.450e+03	1.542e+04 $\approx$	3.134e+03	1.661e+04 $\approx$	2.337e+03
f_{20}	2.168e+03	1.483e+02	2.083e+03 $\approx$	1.017e+02	2.184e+03 $\approx$	1.367e+02	2.136e+03 $\approx$	1.201e+02
f_{21}	2.323e+03	4.516e+00	2.422e+03 $\approx$	7.273e+01	2.326e+03 $\approx$	4.526e+00	2.322e+03 $\approx$	5.120e+00
f_{22}	2.831e+03	1.088e+03	2.474e+03 $\approx$	2.080e+03	2.531e+03 $\approx$	6.940e+02	2.300e+03 $-$	1.233e-03
f_{23}	2.758e+03	1.238e+01	2.738e+03 $-$	1.085e+01	2.767e+03 $\approx$	9.549e+00	2.748e+03 $-$	1.123e+01
f_{24}	2.923e+03	7.880e+00	2.912e+03 $-$	4.435e+00	2.925e+03 $\approx$	7.227e+00	2.912e+03 $-$	9.542e+00
f_{25}	3.083e+03	1.765e+01	3.055e+03 $-$	2.152e+01	3.090e+03 $\approx$	1.572e+01	3.096e+03 $\approx$	1.546e+01
f_{26}	4.053e+03	1.525e+02	3.746e+03 $-$	7.345e+01	4.153e+03 $+$	1.627e+02	4.131e+03 $+$	4.537e+02
f_{27}	3.345e+03	3.223e+01	3.298e+03 $-$	1.759e+01	3.362e+03 $\approx$	2.990e+01	3.333e+03 $\approx$	1.980e+01
f_{28}	3.359e+03	2.528e+01	3.333e+03 $-$	2.459e+01	3.366e+03 $\approx$	1.749e+01	3.381e+03 $\approx$	2.126e+01
f_{29}	3.480e+03	1.702e+02	3.336e+03 $-$	1.369e+02	3.567e+03 $\approx$	1.497e+02	3.516e+03 $\approx$	1.960e+02
f_{30}	9.225e+05	8.296e+04	8.713e+05 $\approx$	6.428e+04	9.137e+05 $\approx$	7.998e+04	9.207e+05 $\approx$	6.919e+04
+/ $\approx$ / $-$			1/17/11		2/24/3		1/20/8	
Avg. ranks	2.9		2.0		2.7		2.4	
Corrected ranks	4		1		3		2	

linear population reduction strategy dynamically adjusts search preferences across different optimization phases. To evaluate SACSO, we conducted comprehensive numerical experiments in CEC2017, CEC2022, and seven classic engineering problems. The experimental results and statistical analysis confirm the competitiveness of SACSO against six state-of-the-art optimizers and four advanced CSO variants. Ablation studies further demonstrate that both strategies introduced in SACSO effectively accelerate optimization convergence. Moreover, we applied SACSO to real-world Human-Powered Aircraft (HPA) design problems, and the results indicate that SACSO is highly promising for solving real-world applications.

Although the proposed SACSO has achieved impressive success in many optimization challenges, it also has limitations, such as higher computational complexity than the original CSO and relatively poor performance against the iL-SHADE in CEC2022. In future research, we will continue to develop efficient variants of CSO and adopt them to other research domains such as intelligent transportation and medicine.

Table 16 Results of ablation experiments in 10-D CEC2022

Func	CSO		SCSO		LCSO		SACSO	
	mean	std	mean	std	mean	std	mean	std
f_1	3.012e+02	4.503e-01	3.006e+02 –	2.059e-01	3.002e+02 –	6.021e-02	3.000e+02 –	6.427e-07
f_2	4.075e+02	5.323e+00	4.085e+02 ≈	1.387e+01	4.052e+02 ≈	6.704e+00	4.001e+02 –	8.685e-02
f_3	6.000e+02	3.938e-07	6.000e+02 –	1.898e-07	6.000e+02 –	5.964e-08	6.000e+02 –	7.213e-13
f_4	8.004e+02	2.285e-01	8.003e+02 ≈	1.489e-01	8.002e+02 –	1.585e-01	8.001e+02 –	9.425e-02
f_5	9.000e+02	3.372e-05	9.000e+02 –	1.918e-05	9.000e+02 –	1.867e-05	9.000e+02 –	2.256e-07
f_6	1.935e+04	4.815e+03	1.686e+04 –	5.213e+03	1.408e+04 –	3.218e+03	8.467e+03 –	2.089e+03
f_7	2.031e+03	2.101e+00	2.031e+03 ≈	1.860e+00	2.029e+03 –	2.474e+00	2.026e+03 –	1.972e+00
f_8	2.259e+03	3.970e+01	2.255e+03 ≈	2.528e+01	2.253e+03 ≈	2.630e+01	2.234e+03 –	1.333e+01
f_9	2.300e+03	2.997e-01	2.301e+03 ≈	1.097e-01	2.301e+03 ≈	8.873e-01	2.300e+03 –	8.890e-05
f_{10}	2.611e+03	3.342e+01	2.613e+03 ≈	3.557e+01	2.609e+03 ≈	3.018e+01	2.599e+03 –	3.343e-01
f_{11}	2.600e+03	2.389e-02	2.600e+03 –	5.935e-02	2.600e+03 –	1.221e-02	2.600e+03 –	3.875e-05
f_{12}	2.867e+03	5.744e-01	2.867e+03 ≈	1.750e-01	2.867e+03 ≈	2.559e-01	2.867e+03 –	2.800e-01
+ / ≈ / –	–		0/7/5		0/5/7		0/0/12	
Avg. ranks	3.6		3.2		2.2		1.0	
Corrected ranks	4		3		2		1	

Table 17 Results of ablation experiments in 20-D CEC2022

Func	CSO		SCSO		LCSO		SACSO	
	mean	std	mean	std	mean	std	mean	std
f_1	3.003e+02	1.589e-01	3.001e+02 –	6.779e-02	3.000e+02 –	1.226e-02	3.000e+02 –	3.361e-07
f_2	4.523e+02	7.169e+00	4.523e+02 ≈	7.923e+00	4.522e+02 ≈	7.300e+00	4.517e+02 ≈	5.362e+00
f_3	6.000e+02	7.294e-09	6.000e+02 –	1.682e-09	6.000e+02 –	2.304e-10	6.000e+02 –	4.687e-13
f_4	8.004e+02	1.159e-01	8.003e+02 ≈	8.733e-02	8.003e+02 –	7.472e-02	8.003e+02 –	8.498e-02
f_5	9.000e+02	1.613e-05	9.000e+02 –	1.568e-05	9.000e+02 –	1.434e-05	9.000e+02 –	3.646e-06
f_6	8.526e+04	3.407e+04	5.003e+04 –	1.221e+04	2.865e+04 –	4.502e+03	2.506e+04 –	4.070e+03
f_7	2.036e+03	2.163e+00	2.034e+03 –	1.855e+00	2.031e+03 –	1.688e+00	2.023e+03 –	1.097e+00
f_8	3.579e+03	4.894e+02	3.342e+03 ≈	3.308e+02	3.336e+03 ≈	3.109e+02	2.884e+03 –	2.204e+02
f_9	2.651e+03	4.678e+00	2.651e+03 ≈	6.516e+00	2.654e+03 ≈	6.209e+00	2.647e+03 –	2.295e+00
f_{10}	2.787e+03	4.752e+01	2.787e+03 ≈	4.789e+01	2.805e+03 ≈	5.463e+01	2.793e+03 ≈	5.028e+01
f_{11}	2.600e+03	5.001e-03	2.600e+03 –	6.924e-03	2.600e+03 –	6.217e-03	2.600e+03 –	3.090e-05
f_{12}	2.948e+03	7.469e+00	2.950e+03 ≈	8.323e+00	2.952e+03 ≈	6.784e+00	2.952e+03 ≈	9.180e+00
+ / ≈ / –	–		0/6/6		0/5/7		0/2/10	
Avg. ranks	3.2		2.6		2.9		1.3	
Corrected ranks	4		2		3		1	

A CEC2017 and CEC2022 benchmark functions

Tables 18 and 19 summarize the detailed information of CEC2017 and CEC2022 benchmark functions.

Table 18 Summary of the CEC2017 benchmark functions: Uni.=Unimodal function, Multi.=Simple multimodal function, Hybrid.=Hybrid function, Comp.=Composition function

No.	Func.	Feature	Optimum
f_1	Shifted and Rotated Bent Cigar function	Uni.	100
f_3	Shifted and Rotated Rosenbrock's function	Multi.	300
f_4	Shifted and Rotated Rastrigin's function		400
f_5	Shifted and Rotated Expanded Scaffer's F6 function		500
f_6	Shifted and Rotated Lunacek Bi_Rastrigin function		600
f_7	Shifted and Rotated Non-Continuous Rastrigin's function		700
f_8	Shifted and Rotated Levy function	Hybrid.	800
f_9	Shifted and Rotated Schwefel's function		900
f_{10}	Hybrid function 1 (N = 3)		1000
f_{11}	Hybrid function 2 (N = 3)		1100
f_{12}	Hybrid function 3 (N = 3)		1200
f_{13}	Hybrid function 4 (N = 4)		1300
f_{14}	Hybrid function 5 (N = 4)		1400
f_{15}	Hybrid function 6 (N = 4)		1500
f_{16}	Hybrid function 6 (N = 5)		1600
f_{17}	Hybrid function 6 (N = 5)		1700
f_{18}	Hybrid function 6 (N = 5)	Comp.	1800
f_{19}	Hybrid function (N = 6)		1900
f_{20}	Composition function 1 (N = 3)		2000
f_{21}	Composition function 2 (N = 3)		2100
f_{22}	Composition function 3 (N = 4)		2200
f_{23}	Composition function 4 (N = 4)		2300
f_{24}	Composition function 5 (N = 5)		2400
f_{25}	Composition function 6 (N = 5)		2500
f_{26}	Composition function 7 (N = 6)		2600
f_{27}	Composition function 8 (N = 6)		2700
f_{28}	Composition function 9 (N = 3)		2800
f_{29}	Composition function 10 (N = 3)		2900
f_{30}	Composition function 11 (N = 3)		3000

Search range: $[-100, 100]^D$ **Table 19** Summary of the CEC2022 benchmark functions: Uni.=Unimodal function, Basic.=Basic function, Hybrid.=Hybrid function, Comp.=Composition function

No.	Func.	Feature	Optimum
f_1	Shifted and full Rotated Zakharov Function	Uni.	300
f_2	Shifted and full Rotated Rosenbrock's Function	Basic.	400
f_3	Shifted and full Rotated Expanded Schaffer's f_6 Function		600
f_4	Shifted and full Rotated Non-Continuous Rastrigin's Function		800
f_5	Shifted and full Rotated Levy Function		900
f_6	Hybrid function 1 (N = 3)	Hybrid.	1800
f_7	Hybrid function 2 (N = 6)		2000
f_8	Hybrid function 3 (N = 5)		2200
f_9	Composition function 1 (N = 5)	Comp.	2300
f_{10}	Composition function 2 (N = 4)		2400
f_{11}	Composition function 3 (N = 5)		2600
f_{12}	Composition function 3 (N = 6)		2700

Search range: $[-100, 100]^D$

B Detailed results in CEC2017

Tables 20, 21, 22, and 23 summarize the detailed experimental results and statistical analysis in CEC2017.

C Detailed results in CEC2022

Tables 24 and 25 summarize the detailed experimental results and statistical analysis in CEC2022.

Table 20 The detailed experimental results in 30-D CEC2017

Func.	L-SHADE	CSO	iL-SHADE	MPEDA	jSO	L-SHADE-cnEpSin	MCSO	OLCSO	TPCSO	DEA ² H ²	SACSO
f_1	mean	3.971e+03	2.075e+03	2.399e+03	4.699e+08	1.952e+03	1.812e+03	2.232e+03	1.345e+03	4.129e+03	2.321e+03
	std	3.541e+03	1.918e+03	2.189e+03	1.039e+08	1.390e+03	1.566e+03	1.726e+03	1.149e+03	4.894e+03	1.606e+03
f_3	mean	1.232e+04	4.612e+04	1.537e+03	1.572e+05	1.781e+04	1.687e+04	6.133e+04	5.038e+04	5.001e+02	2.922e+04
	std	2.961e+04	1.006e+04	1.019e+03	1.964e+04	3.497e+03	4.405e+03	8.693e+03	9.849e+03	1.726e+02	5.283e+03
f_4	mean	4.962e+02	5.142e+02	4.946e+02	6.111e+02	5.041e+02	5.130e+02	5.165e+02	5.165e+02	4.877e+02	5.122e+02
	std	1.345e+01	6.198e+00	1.110e+01	2.127e+01	1.265e+01	6.589e+00	5.595e+00	2.566e+00	2.660e+01	9.278e+00
f_5	mean	5.944e+02	5.173e+02	6.233e+02	7.233e+02	6.677e+02	5.178e+02	5.205e+02	5.133e+02	6.373e+02	5.124e+02
	std	1.210e+01	1.234e+01	1.177e+01	1.395e+01	1.261e+01	5.131e+00	4.398e+00	2.736e+00	2.512e+01	2.162e+00
f_6	mean	6.001e+02	6.000e+02	6.000e+02	6.209e+02	6.000e+02	6.000e+02	6.000e+02	6.000e+02	6.240e+02	6.000e+02
	std	3.007e-02	2.024e-03	1.442e-02	2.001e+00	4.984e-03	1.810e-03	4.652e-05	3.188e-04	5.319e+00	1.319e-04
f_7	mean	8.400e+02	8.645e+02	8.602e+02	9.765e+02	8.983e+02	7.525e+02	7.379e+02	7.445e+02	9.650e+02	7.384e+02
	std	1.419e+01	2.162e+01	1.200e+01	1.900e+01	1.124e+01	8.292e+00	1.547e+00	7.245e+00	5.857e+01	2.223e+00
f_8	mean	8.961e+02	8.168e+02	9.206e+02	1.027e+03	9.664e+02	8.187e+02	8.171e+02	8.128e+02	9.300e+02	8.123e+02
	std	1.392e+01	1.114e+01	1.218e+01	1.144e+01	9.948e+00	4.344e+00	3.493e+00	2.835e+00	3.171e+01	2.637e+00
f_9	mean	9.003e+02	9.000e+02	9.001e+02	2.475e+03	9.000e+02	9.000e+02	9.000e+02	9.000e+02	2.787e+03	9.000e+02
	std	2.866e-01	9.897e-05	1.592e-01	4.122e+02	2.156e-02	1.951e-02	1.951e-02	9.901e-02	7.527e+02	1.275e-07
f_{10}	mean	6.942e+03	5.084e+03	7.362e+03	8.161e+03	8.234e+03	3.174e+03	3.090e+03	2.864e+03	5.015e+03	2.349e+03
	std	6.420e+02	6.682e+02	3.652e+02	3.296e+02	5.231e+02	1.191e+03	4.364e+02	5.060e+02	5.771e+02	4.143e+02
f_{11}	mean	1.881e+03	1.205e+03	1.166e+03	3.175e+03	1.184e+03	1.155e+03	1.266e+03	1.214e+03	1.262e+03	1.176e+03
	std	1.664e+03	3.480e+01	3.149e+01	6.258e+02	2.473e+01	2.832e+01	6.770e+01	7.288e+01	7.198e+01	3.403e+01
f_{12}	mean	3.024e+04	8.192e+05	3.032e+04	1.104e+08	3.638e+05	5.356e+05	8.682e+05	8.801e+05	9.613e+04	7.841e+05
	std	1.764e+04	5.249e+05	2.643e+04	2.903e+07	2.010e+05	2.984e+05	3.852e+05	3.174e+05	1.150e+05	1.896e+05
f_{13}	mean	1.843e+08	1.265e+04	1.714e+03	3.120e+07	1.510e+04	1.219e+04	1.396e+04	1.271e+04	1.999e+04	1.147e+04
	std	5.539e+08	4.571e+03	1.842e+02	1.284e+07	6.802e+03	3.651e+03	3.399e+03	5.194e+03	1.788e+04	2.592e+03
f_{14}	mean	3.783e+04	1.502e+05	1.450e+03	2.238e+05	1.526e+03	1.056e+04	3.446e+05	2.672e+05	1.622e+03	1.107e+05
	std	1.585e+05	7.358e+04	6.189e+00	1.673e+05	1.921e+01	9.047e+03	1.683e+05	1.651e+05	8.566e+01	6.670e+04
f_{15}	mean	1.652e+03	2.281e+03	1.564e+03	4.789e+06	1.890e+03	2.700e+03	2.409e+03	2.398e+03	1.328e+04	2.022e+03
	std	3.991e+01	1.107e+03	3.066e+01	2.130e+06	1.245e+02	9.477e+02	8.498e+02	9.689e+02	1.110e+04	4.738e+02
f_{16}	mean	2.552e+03	1.913e+03	2.532e+03	3.461e+03	3.019e+03	1.956e+03	2.122e+03	2.064e+03	2.605e+03	1.862e+03
	std	3.146e+02	2.365e+02	1.925e+02	1.106e+02	2.355e+02	1.785e+02	1.748e+02	1.709e+02	3.654e+02	1.926e+02

Table 21 The detailed experimental results in 30-D CEC2017 (continued)

Func.	L-SHADE	CSO	iL-SHADE	MPEDA	jSO	L-SHADE-cnEpSin	MCSO	OLCSO	TPCSO	DEA ² H ²	SACSO
f_{17} mean	2.035e+03	1.820e+03	1.896e+03	2.421e+03	2.077e+03	2.036e+03	1.786e+03	1.960e+03	1.816e+03	2.365e+03	1.808e+03
f_{17} std	2.560e+02	9.878e+01	4.498e+01	1.148e+02	7.325e+01	8.903e+01	6.010e+01	1.522e+02	8.923e+01	2.941e+02	1.051e+02
f_{18} mean	2.099e+03	3.566e+05	1.922e+03	5.049e+06	2.426e+04	1.149e+04	2.861e+05	4.141e+05	3.147e+05	3.868e+04	2.796e+05
f_{18} std	4.218e+02	1.825e+05	3.957e+01	2.322e+06	8.357e+03	4.139e+03	2.289e+05	4.323e+05	2.285e+05	3.013e+04	1.495e+05
f_{19} mean	1.998e+07	5.600e+03	1.927e+03	6.517e+06	2.016e+03	2.033e+03	5.020e+03	5.272e+03	5.430e+03	5.520e+03	5.303e+03
f_{19} std	8.707e+07	2.231e+03	4.439e+00	3.176e+06	5.064e+01	6.404e+01	2.022e+03	2.562e+03	2.527e+03	4.849e+03	1.145e+03
f_{20} mean	2.481e+03	2.154e+03	2.302e+03	2.734e+03	2.440e+03	2.456e+03	2.144e+03	2.236e+03	2.208e+03	2.526e+03	2.154e+03
f_{20} std	2.184e+02	3.956e+00	8.557e+01	1.402e+02	1.117e+02	9.643e+01	3.619e+01	9.316e+01	1.017e+02	1.616e+02	6.328e+00
f_{21} mean	2.417e+03	2.314e+03	2.407e+03	2.523e+03	2.461e+03	2.431e+03	2.319e+03	2.319e+03	2.313e+03	2.424e+03	2.310e+03
f_{21} std	7.571e+01	5.098e+00	1.802e+01	1.146e+01	9.522e+00	1.444e+01	4.351e+00	6.383e+00	3.827e+00	2.967e+01	3.459e+00
f_{22} mean	5.009e+03	2.300e+03	2.300e+03	8.325e+03	2.300e+03	2.306e+03	2.300e+03	2.300e+03	2.300e+03	4.384e+03	2.300e+03
f_{22} std	2.798e+03	3.219e-03	7.143e-01	1.204e+03	8.263e-02	2.564e+00	4.315e-03	6.033e-05	7.954e-04	2.120e+03	9.327e-04
f_{23} mean	2.861e+03	2.663e+03	2.750e+03	2.883e+03	2.805e+03	2.789e+03	2.663e+03	2.680e+03	2.672e+03	2.855e+03	2.662e+03
f_{23} std	1.364e+02	6.286e+00	2.146e+01	1.298e+01	1.697e+01	1.415e+01	7.016e+00	9.803e+00	7.226e+00	4.570e+01	6.943e+00
f_{24} mean	2.990e+03	2.828e+03	2.923e+03	3.052e+03	2.973e+03	2.950e+03	2.835e+03	2.836e+03	2.834e+03	2.995e+03	2.824e+03
f_{24} std	1.138e+02	3.075e+00	1.993e+01	1.101e+01	9.819e+00	1.380e+01	4.309e+00	7.354e+00	7.000e+00	4.536e+01	3.553e+00
f_{25} mean	2.887e+03	2.888e+03	2.887e+03	2.955e+03	2.887e+03	2.889e+03	2.888e+03	2.902e+03	2.890e+03	2.892e+03	2.893e+03
f_{25} std	2.342e-01	2.283e+00	2.452e-01	1.103e+01	4.353e-01	1.705e+00	3.671e+00	1.325e+01	3.400e+00	8.802e+00	9.814e+00
f_{26} mean	4.723e+03	3.693e+03	4.383e+03	6.011e+03	4.975e+03	4.855e+03	3.544e+03	4.070e+03	3.857e+03	5.487e+03	3.522e+03
f_{26} std	1.047e+03	2.209e+02	1.821e+02	1.039e+02	1.577e+02	1.642e+02	3.068e+02	6.172e+02	3.128e+02	1.201e+03	4.227e+02
f_{27} mean	3.243e+03	3.219e+03	3.205e+03	3.278e+03	3.207e+03	3.219e+03	3.211e+03	3.233e+03	3.228e+03	3.264e+03	3.220e+03
f_{27} std	5.426e+01	6.819e+00	1.082e+01	1.113e+01	6.571e+00	7.189e+00	6.852e+00	9.541e+00	9.231e+00	3.098e+01	7.162e+00
f_{28} mean	3.319e+03	3.234e+03	3.221e+03	3.400e+03	3.229e+03	3.230e+03	3.226e+03	3.274e+03	3.258e+03	3.206e+03	3.239e+03
f_{28} std	4.353e+02	1.283e+01	1.558e+01	2.959e+01	1.214e+01	1.492e+01	1.486e+01	1.752e+01	1.531e+01	4.522e+01	1.734e+01
f_{29} mean	3.741e+03	3.460e+03	3.671e+03	4.536e+03	3.857e+03	3.790e+03	3.371e+03	3.524e+03	3.439e+03	4.186e+03	3.407e+03
f_{29} std	3.961e+02	1.029e+02	8.919e+01	1.270e+02	1.123e+02	8.687e+01	2.213e+01	1.046e+02	1.099e+02	2.777e+02	7.813e+01
f_{30} mean	1.097e+07	8.290e+03	6.435e+03	6.394e+06	1.784e+04	1.628e+04	7.823e+03	1.748e+04	9.595e+03	1.194e+04	8.075e+03
f_{30} std	4.776e+07	1.137e+03	6.211e+02	2.112e+06	7.364e+03	5.059e+03	9.779e+02	7.931e+03	1.916e+03	5.071e+03	1.146e+03
+/ $\approx$ /-;	19/2/8	13/16/0	15/2/12	29/0/0	16/7/6	17/7/5	11/16/2	19/10/0	19/10/0	19/4/6	-
Avg. ranks:	6.9	5.0	4.2	10.7	6.3	6.6	3.8	6.0	5.1	7.6	3.4
Corrected ranks:	9	4	3	11	7	8	2	6	5	10	1

Table 22 The detailed experimental results in 50-D CEC2017

Func.	L-SHADE	CSO	iL-SHADE	MPEDA	jSO	L-SHADE-cnEpSin	MCSO	OLCSO	TPCSO	DEA ² H ²	SACSO
f_1	mean	1.332e+06 +	1.205e+03 ≈	5.847e+04 +	1.187e+08 +	2.967e+04 +	5.905e+06 +	1.106e+03 ≈	1.099e+03 ≈	2.338e+04 +	1.087e+03
	std	7.741e+05	1.411e+03	4.972e+04	2.737e+07	2.796e+04	2.614e+06	6.883e+02	1.092e+03	5.297e+04	7.179e+02
f_3	mean	1.120e+05 +	1.028e+05 +	1.377e+04 −	2.944e+05 +	7.768e+04 +	8.291e+04 +	1.358e+05 +	1.178e+05 +	3.991e+03 −	7.058e+04
	std	1.352e+05	1.279e+04	3.898e+03	3.373e+04	7.513e+03	2.031e+04	1.255e+04	1.794e+04	1.316e+03	1.240e+04
f_4	mean	5.819e+02 ≈	5.736e+02 ≈	5.440e+02 −	6.836e+02 +	5.793e+02 ≈	5.924e+02 ≈	6.212e+02 +	5.899e+02 ≈	5.566e+02 −	5.996e+02
	std	3.338e+01	3.137e+01	4.422e+01	1.736e+01	2.981e+01	3.139e+01	2.528e+01	2.878e+01	3.645e+01	3.065e+01
f_5	mean	7.200e+02 +	5.237e+02 ≈	7.534e+02 +	8.966e+02 +	8.336e+02 +	7.818e+02 +	5.498e+02 +	5.322e+02 +	7.897e+02 +	5.266e+02
	std	1.931e+01	4.451e+00	2.101e+01	1.578e+01	1.544e+01	1.932e+01	6.129e+00	4.948e+00	6.301e+01	2.933e+00
f_6	mean	6.003e+02 +	6.000e+02 +	6.002e+02 +	6.140e+02 +	6.000e+02 +	6.014e+02 +	6.000e+02 +	6.000e+02 ≈	6.372e+02 +	6.000e+02
	std	1.173e−01	1.782e−04	8.424e−02	2.254e+00	4.988e−03	3.626e−01	2.223e−02	1.940e−05	6.266e+00	8.827e−05
f_7	mean	9.790e+02 +	8.217e+02 +	1.022e+03 +	1.161e+03 +	1.079e+03 +	1.050e+03 +	7.698e+02 +	7.648e+02 ≈	1.321e+03 +	7.649e+02
	std	1.944e+01	4.548e+01	1.366e+01	1.706e+01	1.384e+01	1.823e+01	4.468e+00	2.807e+00	1.503e+02	2.495e+00
f_8	mean	1.039e+03 +	8.229e+02 ≈	1.062e+03 +	1.193e+03 +	1.129e+03 +	1.084e+03 +	8.501e+02 +	8.331e+02 +	1.073e+03 +	8.255e+02
	std	1.734e+01	2.952e+00	2.121e+01	1.686e+01	1.656e+01	2.135e+01	7.017e+00	6.100e+00	5.552e+01	4.212e+00
f_9	mean	9.106e+02 +	9.003e+02 ≈	9.046e+02 +	1.989e+03 +	9.010e+02 +	9.298e+02 +	9.227e+02 +	9.008e+02 +	8.605e+03 +	9.003e+02
	std	5.108e+00	2.741e−01	2.239e+00	3.584e+02	5.926e−01	1.414e+01	1.588e+01	1.178e+00	2.523e+03	3.755e−01
f_{10}	mean	1.266e+04 +	3.208e+03 ≈	1.303e+04 +	1.395e+04 +	1.439e+04 +	1.318e+04 +	4.749e+03 +	3.857e+03 +	7.941e+03 +	3.110e+03
	std	8.306e+02	6.819e+02	4.066e+02	3.151e+02	4.679e+02	1.045e+03	6.475e+02	5.216e+02	9.943e+02	4.089e+02
f_{11}	mean	1.241e+03 −	1.919e+03 +	1.219e+03 −	8.788e+03 +	1.309e+03 −	1.320e+03 −	2.633e+03 +	2.102e+03 +	1.373e+03 −	1.783e+03
	std	1.847e+01	3.839e+02	3.017e+01	5.407e+03	3.099e+01	3.956e+01	9.046e+02	4.283e+02	7.649e+01	2.706e+02
f_{12}	mean	1.510e+06 −	2.367e+06 ≈	1.137e+06 −	1.035e+08 +	2.076e+06 ≈	3.430e+06 +	4.445e+06 ≈	2.540e+06 ≈	1.034e+06 −	2.206e+06
	std	4.212e+05	9.006e+05	4.570e+05	2.846e+07	6.749e+05	1.041e+06	2.090e+06	1.122e+06	9.080e+05	5.997e+05
f_{13}	mean	1.223e+09 +	2.458e+03 ≈	5.238e+03 +	2.140e+07 +	2.593e+04 +	3.678e+04 +	7.588e+03 +	2.699e+03 +	1.122e+04 +	2.420e+03
	std	3.699e+09	1.059e+03	1.006e+03	8.173e+06	9.256e+03	8.719e+03	2.151e+03	1.134e+03	6.578e+03	6.385e+02
f_{14}	mean	1.573e+03 −	3.197e+05 ≈	1.545e+03 −	6.376e+05 ≈	2.639e+03 −	1.763e+03 −	7.933e+05 +	5.415e+05 +	4.674e+03 −	3.666e+05
	std	2.769e+01	1.738e+05	1.754e+01	5.768e+05	7.620e+02	1.017e+02	3.929e+05	2.714e+05	2.349e+03	2.205e+05
f_{15}	mean	2.175e+03 −	4.980e+03 ≈	1.941e+03 −	6.243e+06 +	5.283e+03 ≈	5.907e+03 ≈	6.191e+03 ≈	5.472e+03 ≈	1.099e+04 +	5.498e+03
	std	1.301e+02	2.195e+03	7.543e+01	2.749e+06	1.122e+03	1.354e+03	1.947e+03	1.992e+03	1.027e+04	1.606e+03
f_{16}	mean	3.473e+03 +	2.194e+03 ≈	3.478e+03 +	5.164e+03 +	4.426e+03 +	3.953e+03 +	2.324e+03 +	2.246e+03 +	3.603e+03 +	2.138e+03
	std	6.934e+02	1.697e+02	2.407e+02	1.437e+02	2.391e+02	3.322e+02	2.107e+02	2.376e+02	4.603e+02	1.413e+02

Table 23 The detailed experimental results in 50-D CEC2017 (continued)

Func.	L-SHADE	CSO	iL-SHADE	MPEDA	jSO	L-SHADE-cnEpSin	MCSO	OLCSO	TPCSO	DEA ² H ²	SACSO
f_{17} mean	2.887e+03 + 2.061e+03	2.932e+03 + 3.859e+03 + 3.494e+03 + 3.356e+03 +	2.932e+03 + 3.859e+03 + 3.494e+03 + 3.356e+03 +	2.932e+03 + 3.859e+03 + 3.494e+03 + 3.356e+03 +	2.932e+03 + 3.859e+03 + 3.494e+03 + 3.356e+03 +	2.932e+03 + 3.859e+03 + 3.494e+03 + 3.356e+03 +	2.122e+03 + 2.487e+03 + 2.206e+03 + 3.326e+03 + 2.147e+03	2.122e+03 + 2.487e+03 + 2.206e+03 + 3.326e+03 + 2.147e+03	2.122e+03 + 2.487e+03 + 2.206e+03 + 3.326e+03 + 2.147e+03	2.122e+03 + 2.487e+03 + 2.206e+03 + 3.326e+03 + 2.147e+03	2.122e+03 + 2.487e+03 + 2.206e+03 + 3.326e+03 + 2.147e+03
f_{17} std	1.935e+02	1.802e+02	2.351e+02	2.310e+02	1.809e+02	2.180e+02	2.438e+02	2.099e+02	2.298e+02	3.885e+02	1.940e+02
f_{18} mean	3.853e+03 - 2.574e+03	2.137e+06 + 8.839e+04 - 3.992e+04 -	2.137e+06 + 8.839e+04 - 3.992e+04 -	2.137e+06 + 8.839e+04 - 3.992e+04 -	2.137e+06 + 8.839e+04 - 3.992e+04 -	2.137e+06 + 8.839e+04 - 3.992e+04 -	1.043e+06 + 2.280e+06 + 6.133e+05 + 3.618e+04 + 4.040e+05	1.043e+06 + 2.280e+06 + 6.133e+05 + 3.618e+04 + 4.040e+05	1.043e+06 + 2.280e+06 + 6.133e+05 + 3.618e+04 + 4.040e+05	1.043e+06 + 2.280e+06 + 6.133e+05 + 3.618e+04 + 4.040e+05	1.043e+06 + 2.280e+06 + 6.133e+05 + 3.618e+04 + 4.040e+05
f_{18} std	4.096e+03	6.512e+05	4.922e+06	4.922e+06	2.874e+04	1.505e+04	3.820e+05	6.362e+05	6.133e+05	3.618e+04	4.040e+05
f_{19} mean	2.027e+03 - 1.982e+03	1.567e+04 + 9.673e+03 - 9.278e+03 -	1.567e+04 + 9.673e+03 - 9.278e+03 -	1.567e+04 + 9.673e+03 - 9.278e+03 -	1.567e+04 + 9.673e+03 - 9.278e+03 -	1.567e+04 + 9.673e+03 - 9.278e+03 -	1.470e+04 + 1.621e+04 + 1.474e+04 + 1.630e+04 + 1.661e+04	1.470e+04 + 1.621e+04 + 1.474e+04 + 1.630e+04 + 1.661e+04	1.470e+04 + 1.621e+04 + 1.474e+04 + 1.630e+04 + 1.661e+04	1.470e+04 + 1.621e+04 + 1.474e+04 + 1.630e+04 + 1.661e+04	1.470e+04 + 1.621e+04 + 1.474e+04 + 1.630e+04 + 1.661e+04
f_{19} std	3.261e+01	3.467e+03	1.727e+01	1.171e+06	3.447e+03	4.076e+03	5.018e+03	3.912e+03	2.624e+03	9.871e+03	2.337e+03
f_{20} mean	3.338e+03 + 2.168e+03 + 3.073e+03 + 4.000e+03 + 3.739e+03 + 3.510e+03 +	3.338e+03 + 2.168e+03 + 3.073e+03 + 4.000e+03 + 3.739e+03 + 3.510e+03 +	3.338e+03 + 2.168e+03 + 3.073e+03 + 4.000e+03 + 3.739e+03 + 3.510e+03 +	3.338e+03 + 2.168e+03 + 3.073e+03 + 4.000e+03 + 3.739e+03 + 3.510e+03 +	3.338e+03 + 2.168e+03 + 3.073e+03 + 4.000e+03 + 3.739e+03 + 3.510e+03 +	3.338e+03 + 2.168e+03 + 3.073e+03 + 4.000e+03 + 3.739e+03 + 3.510e+03 +	2.190e+03 + 2.439e+03 + 2.284e+03 + 3.425e+03 + 2.136e+03	2.190e+03 + 2.439e+03 + 2.284e+03 + 3.425e+03 + 2.136e+03	2.190e+03 + 2.439e+03 + 2.284e+03 + 3.425e+03 + 2.136e+03	2.190e+03 + 2.439e+03 + 2.284e+03 + 3.425e+03 + 2.136e+03	2.190e+03 + 2.439e+03 + 2.284e+03 + 3.425e+03 + 2.136e+03
f_{20} std	2.821e+02	1.483e+02	1.400e+02	1.724e+02	1.845e+02	1.660e+02	1.278e+02	1.558e+02	1.897e+02	2.806e+02	1.201e+02
f_{21} mean	2.562e+03 + 2.323e+03 + 2.551e+03 + 2.700e+03 + 2.622e+03 + 2.583e+03 +	2.562e+03 + 2.323e+03 + 2.551e+03 + 2.700e+03 + 2.622e+03 + 2.583e+03 +	2.562e+03 + 2.323e+03 + 2.551e+03 + 2.700e+03 + 2.622e+03 + 2.583e+03 +	2.562e+03 + 2.323e+03 + 2.551e+03 + 2.700e+03 + 2.622e+03 + 2.583e+03 +	2.562e+03 + 2.323e+03 + 2.551e+03 + 2.700e+03 + 2.622e+03 + 2.583e+03 +	2.562e+03 + 2.323e+03 + 2.551e+03 + 2.700e+03 + 2.622e+03 + 2.583e+03 +	2.337e+03 + 2.342e+03 + 2.329e+03 + 2.600e+03 + 2.322e+03	2.337e+03 + 2.342e+03 + 2.329e+03 + 2.600e+03 + 2.322e+03	2.337e+03 + 2.342e+03 + 2.329e+03 + 2.600e+03 + 2.322e+03	2.337e+03 + 2.342e+03 + 2.329e+03 + 2.600e+03 + 2.322e+03	2.337e+03 + 2.342e+03 + 2.329e+03 + 2.600e+03 + 2.322e+03
f_{21} std	1.442e+02	4.516e+00	2.568e+01	1.419e+01	1.790e+01	2.028e+01	5.933e+00	6.198e+00	6.125e+00	6.572e+01	5.120e+00
f_{22} mean	1.404e+04 + 2.831e+03 + 1.328e+04 + 1.536e+04 + 1.444e+04 + 1.285e+04 +	1.404e+04 + 2.831e+03 + 1.328e+04 + 1.536e+04 + 1.444e+04 + 1.285e+04 +	1.404e+04 + 2.831e+03 + 1.328e+04 + 1.536e+04 + 1.444e+04 + 1.285e+04 +	1.404e+04 + 2.831e+03 + 1.328e+04 + 1.536e+04 + 1.444e+04 + 1.285e+04 +	1.404e+04 + 2.831e+03 + 1.328e+04 + 1.536e+04 + 1.444e+04 + 1.285e+04 +	1.404e+04 + 2.831e+03 + 1.328e+04 + 1.536e+04 + 1.444e+04 + 1.285e+04 +	3.855e+03 + 2.849e+03 + 3.017e+03 + 9.940e+03 + 2.300e+03	3.855e+03 + 2.849e+03 + 3.017e+03 + 9.940e+03 + 2.300e+03	3.855e+03 + 2.849e+03 + 3.017e+03 + 9.940e+03 + 2.300e+03	3.855e+03 + 2.849e+03 + 3.017e+03 + 9.940e+03 + 2.300e+03	3.855e+03 + 2.849e+03 + 3.017e+03 + 9.940e+03 + 2.300e+03
f_{22} std	1.771e+03	1.088e+03	3.278e+03	6.079e+02	3.950e+03	2.975e+03	1.578e+03	1.536e+03	1.453e+03	1.129e+03	1.233e+03
f_{23} mean	3.118e+03 + 2.758e+03 + 2.980e+03 + 3.152e+03 + 3.051e+03 + 3.010e+03 +	3.118e+03 + 2.758e+03 + 2.980e+03 + 3.152e+03 + 3.051e+03 + 3.010e+03 +	3.118e+03 + 2.758e+03 + 2.980e+03 + 3.152e+03 + 3.051e+03 + 3.010e+03 +	3.118e+03 + 2.758e+03 + 2.980e+03 + 3.152e+03 + 3.051e+03 + 3.010e+03 +	3.118e+03 + 2.758e+03 + 2.980e+03 + 3.152e+03 + 3.051e+03 + 3.010e+03 +	3.118e+03 + 2.758e+03 + 2.980e+03 + 3.152e+03 + 3.051e+03 + 3.010e+03 +	2.751e+03 + 2.801e+03 + 2.771e+03 + 3.184e+03 + 2.748e+03	2.751e+03 + 2.801e+03 + 2.771e+03 + 3.184e+03 + 2.748e+03	2.751e+03 + 2.801e+03 + 2.771e+03 + 3.184e+03 + 2.748e+03	2.751e+03 + 2.801e+03 + 2.771e+03 + 3.184e+03 + 2.748e+03	2.751e+03 + 2.801e+03 + 2.771e+03 + 3.184e+03 + 2.748e+03
f_{23} std	2.540e+02	1.238e+01	1.973e+01	1.590e+01	2.121e+01	2.376e+01	1.275e+01	2.193e+01	1.457e+01	1.320e+02	1.123e+01
f_{24} mean	3.402e+03 + 2.923e+03 + 3.105e+03 + 3.321e+03 + 3.214e+03 + 3.170e+03 +	3.402e+03 + 2.923e+03 + 3.105e+03 + 3.321e+03 + 3.214e+03 + 3.170e+03 +	3.402e+03 + 2.923e+03 + 3.105e+03 + 3.321e+03 + 3.214e+03 + 3.170e+03 +	3.402e+03 + 2.923e+03 + 3.105e+03 + 3.321e+03 + 3.214e+03 + 3.170e+03 +	3.402e+03 + 2.923e+03 + 3.105e+03 + 3.321e+03 + 3.214e+03 + 3.170e+03 +	3.402e+03 + 2.923e+03 + 3.105e+03 + 3.321e+03 + 3.214e+03 + 3.170e+03 +	2.929e+03 + 2.964e+03 + 2.934e+03 + 3.293e+03 + 2.912e+03	2.929e+03 + 2.964e+03 + 2.934e+03 + 3.293e+03 + 2.912e+03	2.929e+03 + 2.964e+03 + 2.934e+03 + 3.293e+03 + 2.912e+03	2.929e+03 + 2.964e+03 + 2.934e+03 + 3.293e+03 + 2.912e+03	2.929e+03 + 2.964e+03 + 2.934e+03 + 3.293e+03 + 2.912e+03
f_{24} std	2.431e+02	7.880e+00	3.259e+01	1.663e+01	1.454e+01	2.616e+01	9.021e+00	1.385e+01	9.683e+00	1.124e+02	9.542e+00
f_{25} mean	3.065e+03 - 3.040e+03	3.083e+03 + 3.129e+03 + 3.055e+03 - 3.074e+03 -	3.083e+03 + 3.129e+03 + 3.055e+03 - 3.074e+03 -	3.083e+03 + 3.129e+03 + 3.055e+03 - 3.074e+03 -	3.083e+03 + 3.129e+03 + 3.055e+03 - 3.074e+03 -	3.083e+03 + 3.129e+03 + 3.055e+03 - 3.074e+03 -	3.056e+03 - 3.139e+03 + 3.105e+03 + 3.071e+03 + 3.096e+03	3.056e+03 - 3.139e+03 + 3.105e+03 + 3.071e+03 + 3.096e+03	3.056e+03 - 3.139e+03 + 3.105e+03 + 3.071e+03 + 3.096e+03	3.056e+03 - 3.139e+03 + 3.105e+03 + 3.071e+03 + 3.096e+03	3.056e+03 - 3.139e+03 + 3.105e+03 + 3.071e+03 + 3.096e+03
f_{25} std	2.365e+01	1.765e+01	1.708e+01	1.444e+01	1.759e+01	2.273e+01	2.175e+01	2.681e+01	1.576e+01	3.927e+01	1.546e+01
f_{26} mean	6.533e+03 + 4.053e+03 - 5.562e+03 + 7.830e+03 + 6.627e+03 + 6.350e+03 +	6.533e+03 + 4.053e+03 - 5.562e+03 + 7.830e+03 + 6.627e+03 + 6.350e+03 +	6.533e+03 + 4.053e+03 - 5.562e+03 + 7.830e+03 + 6.627e+03 + 6.350e+03 +	6.533e+03 + 4.053e+03 - 5.562e+03 + 7.830e+03 + 6.627e+03 + 6.350e+03 +	6.533e+03 + 4.053e+03 - 5.562e+03 + 7.830e+03 + 6.627e+03 + 6.350e+03 +	6.533e+03 + 4.053e+03 - 5.562e+03 + 7.830e+03 + 6.627e+03 + 6.350e+03 +	4.008e+03 - 5.164e+03 + 4.452e+03 + 8.297e+03 + 4.131e+03	4.008e+03 - 5.164e+03 + 4.452e+03 + 8.297e+03 + 4.131e+03	4.008e+03 - 5.164e+03 + 4.452e+03 + 8.297e+03 + 4.131e+03	4.008e+03 - 5.164e+03 + 4.452e+03 + 8.297e+03 + 4.131e+03	4.008e+03 - 5.164e+03 + 4.452e+03 + 8.297e+03 + 4.131e+03
f_{26} std	2.214e+03	1.525e+02	4.411e+02	1.679e+02	1.651e+02	2.253e+02	1.114e+02	6.155e+02	1.991e+02	2.150e+03	4.537e+02
f_{27} mean	3.429e+03 + 3.345e+03 + 3.280e+03 - 3.280e+03 - 3.304e+03 + 3.349e+03 +	3.429e+03 + 3.345e+03 + 3.280e+03 - 3.280e+03 - 3.304e+03 + 3.349e+03 +	3.429e+03 + 3.345e+03 + 3.280e+03 - 3.280e+03 - 3.304e+03 + 3.349e+03 +	3.429e+03 + 3.345e+03 + 3.280e+03 - 3.280e+03 - 3.304e+03 + 3.349e+03 +	3.429e+03 + 3.345e+03 + 3.280e+03 - 3.280e+03 - 3.304e+03 + 3.349e+03 +	3.429e+03 + 3.345e+03 + 3.280e+03 - 3.280e+03 - 3.304e+03 + 3.349e+03 +	3.292e+03 + 3.484e+03 + 3.413e+03 + 3.708e+03 + 3.333e+03	3.292e+03 + 3.484e+03 + 3.413e+03 + 3.708e+03 + 3.333e+03	3.292e+03 + 3.484e+03 + 3.413e+03 + 3.708e+03 + 3.333e+03	3.292e+03 + 3.484e+03 + 3.413e+03 + 3.708e+03 + 3.333e+03	3.292e+03 + 3.484e+03 + 3.413e+03 + 3.708e+03 + 3.333e+03
f_{27} std	1.785e+02	3.223e+01	4.274e+01	7.277e+01	2.953e+01	4.032e+01	2.607e+01	5.071e+01	3.544e+01	1.806e+02	1.980e+01
f_{28} mean	3.407e+03 + 3.359e+03 + 3.302e+03 - 3.302e+03 - 3.351e+03 + 3.369e+03 +	3.407e+03 + 3.359e+03 + 3.302e+03 - 3.302e+03 - 3.351e+03 + 3.369e+03 +	3.407e+03 + 3.359e+03 + 3.302e+03 - 3.302e+03 - 3.351e+03 + 3.369e+03 +	3.407e+03 + 3.359e+03 + 3.302e+03 - 3.302e+03 - 3.351e+03 + 3.369e+03 +	3.407e+03 + 3.359e+03 + 3.302e+03 - 3.302e+03 - 3.351e+03 + 3.369e+03 +	3.407e+03 + 3.359e+03 + 3.302e+03 - 3.302e+03 - 3.351e+03 + 3.369e+03 +	3.329e+03 - 3.526e+03 + 3.408e+03 + 3.313e+03 - 3.381e+03	3.329e+03 - 3.526e+03 + 3.408e+03 + 3.313e+03 - 3.381e+03	3.329e+03 - 3.526e+03 + 3.408e+03 + 3.313e+03 - 3.381e+03	3.329e+03 - 3.526e+03 + 3.408e+03 + 3.313e+03 - 3.381e+03	3.329e+03 - 3.526e+03 + 3.408e+03 + 3.313e+03 - 3.381e+03
f_{28} std	1.392e+02	2.528e+01	3.052e+01	4.614e+01	2.222e+01	3.592e+01	1.920e+01	6.297e+01	2.816e+01	3.320e+01	2.126e+01
f_{29} mean	4.524e+03 + 3.480e+03 + 4.059e+03 + 5.465e+03 + 4.448e+03 + 4.408e+03 +	4.524e+03 + 3.480e+03 + 4.059e+03 + 5.465e+03 + 4.448e+03 + 4.408e+03 +	4.524e+03 + 3.480e+03 + 4.059e+03 + 5.465e+03 + 4.448e+03 + 4.408e+03 +	4.524e+03 + 3.480e+03 + 4.059e+03 + 5.465e+03 + 4.448e+03 + 4.408e+03 +	4.524e+03 + 3.480e+03 + 4.059e+03 + 5.465e+03 + 4.448e+03 + 4.408e+03 +	4.524e+03 + 3.480e+03 + 4.059e+03 + 5.465e+03 + 4.448e+03 + 4.408e+03 +	3.432e+03 + 3.950e+03 + 3.702e+03 + 4.905e+03 + 3.516e+03	3.432e+03 + 3.950e+03 + 3.702e+03 + 4.905e+03 + 3.516e+03	3.432e+03 + 3.950e+03 + 3.702e+03 + 4.905e+03 + 3.516e+03	3.432e+03 + 3.950e+03 + 3.702e+03 + 4.905e+03 + 3.516e+03	3.432e+03 + 3.950e+03 + 3.702e+03 + 4.905e+03 + 3.516e+03
f_{29} std	1.144e+03	1.702e+02	1.851e+02	1.405e+02	2.526e+02	1.942e+02	1.247e+02	1.886e+02	1.887e+02	5.224e+02	1.960e+02
f_{30} mean	1.090e+06 + 9.225e+05 + 9.058e+05 + 5.418e+07 + 3.421e+06 + 3.801e+06 +	1.090e+06 + 9.225e+05 + 9.058e+05 + 5.418e+07 + 3.421e+06 + 3.801e+06 +	1.090e+06 + 9.225e+05 + 9.058e+05 + 5.418e+07 + 3.421e+06 + 3.801e+06 +	1.090e+06 + 9.225e+05 + 9.058e+05 + 5.418e+07 + 3.421e+06 + 3.801e+06 +	1.090e+06 + 9.225e+05 + 9.058e+05 + 5.418e+07 + 3.421e+06 + 3.801e+06 +	1.090e+06 + 9.225e+05 + 9.058e+05 + 5.418e+07 + 3.421e+06 + 3.801e+06 +	8.711e+05 + 3.419e+06 + 1.072e+06 + 1.015e+06 + 9.207e+05	8.711e+05 + 3.419e+06 + 1.072e+06 + 1.015e+06 + 9.207e+05	8.711e+05 + 3.419e+06 + 1.072e+06 + 1.015e+06 + 9.207e+05	8.711e+05 + 3.419e+06 + 1.072e+06 + 1.015e+06 + 9.207e+05	8.711e+05 + 3.419e+06 + 1.072e+06 + 1.015e+06 + 9.207e+05
f_{30} std	1.528e+05	8.296e+04	9.490e+04	1.800e+07	7.069e+05	5.919e+05	5.721e+04	7.960e+05	1.174e+05	2.107e+05	6.919e+04
+/-	19/3/7	8/20/1	17/1/11	28/1/0	19/5/5	20/4/5	10/11/8	25/4/0	19/		

Table 24 The detailed experimental results in 10-D CEC2022

Func.		L-SHADE	CSO	iL-SHADE	MPEDA	jSO	L-SHADE-cnEpSin	MCSO	OLCSO	TPCSO	DEA ² H ²	SACSO
f_1	mean	3.000e+02 –	3.012e+02 +	3.000e+02 –	3.373e+03 +	3.000e+02 +	3.000e+02 +	3.000e+02 +	3.001e+02 +	3.004e+02 +	3.000e+02 +	3.000e+02 –
	std	2.481e–07	4.503e–01	1.296e–11	8.316e+02	8.275e–07	5.654e–04	1.797e–03	1.399e–02	1.682e–01	1.961e–10	6.427e–07
f_2	mean	4.078e+02 +	4.075e+02 +	4.055e+02 +	4.439e+02 +	4.064e+02 +	4.074e+02 +	4.020e+02 +	4.041e+02 +	4.040e+02 +	4.098e+02 +	4.001e+02
	std	1.811e+00	5.323e+00	3.702e+00	1.044e+01	3.723e+00	2.680e+00	3.212e+00	3.890e+00	4.309e+00	1.671e+01	8.685e–02
f_3	mean	6.000e+02 ≈	6.000e+02 +	6.000e+02 –	6.000e+02 +	6.000e+02 –	6.000e+02 +	6.000e+02 +	6.000e+02 +	6.000e+02 +	6.000e+02 +	6.000e+02
	std	3.275e–13	3.938e–07	0.000e+00	3.589e–03	9.166e–14	2.444e–10	1.759e–08	2.372e–08	8.996e–08	2.950e–13	7.213e–13
f_4	mean	8.004e+02 +	8.004e+02 +	8.003e+02 +	8.010e+02 +	8.006e+02 +	8.004e+02 +	8.002e+02 +	8.002e+02 +	8.003e+02 +	8.003e+02 +	8.001e+02
	std	3.746e–01	2.285e–01	1.017e–01	2.125e–01	1.362e–01	1.293e–01	9.276e–02	1.464e–01	1.773e–01	1.335e–01	9.425e–02
f_5	mean	9.000e+02 –	9.000e+02 +	9.000e+02 –	9.007e+02 +	9.000e+02 ≈	9.000e+02 +	9.000e+02 +	9.000e+02 +	9.000e+02 +	9.010e+02 +	9.000e+02
	std	3.045e–10	3.372e–05	2.163e–12	2.571e–01	2.546e–07	1.547e–06	2.348e–06	5.659e–06	3.207e–05	8.263e–01	2.256e–07
f_6	mean	9.292e+04 +	1.935e+04 +	1.801e+03 –	6.900e+05 +	1.923e+03 –	2.435e+03 –	1.712e+04 +	1.541e+04 +	1.461e+04 +	6.466e+03 –	8.467e+03
	std	3.971e+05	4.815e+03	5.124e–01	5.645e+05	2.032e+02	7.168e+02	4.551e+03	3.739e+03	4.464e+03	7.981e+03	2.089e+03
f_7	mean	2.028e+03 +	2.031e+03 +	2.031e+03 +	2.108e+03 +	2.030e+03 +	2.033e+03 +	2.034e+03 +	2.028e+03 +	2.030e+03 +	2.026e+03 ≈	2.026e+03
	std	1.633e+00	2.101e+00	2.770e+00	2.173e+01	4.148e+00	3.936e+00	4.357e+00	2.091e+00	2.619e+00	6.056e+00	1.972e+00
f_8	mean	2.218e+03 –	2.259e+03 +	2.215e+03 –	2.397e+03 +	2.218e+03 –	2.223e+03 –	2.227e+03 ≈	2.253e+03 +	2.256e+03 +	2.221e+03 –	2.234e+03
	std	7.069e+00	3.970e+01	7.721e+00	7.019e+01	7.517e+00	5.429e+00	4.836e+00	2.354e+01	4.526e+01	3.572e+00	1.333e+01
f_9	mean	2.318e+03 +	2.300e+03 +	2.300e+03 –	2.641e+03 +	2.318e+03 +	2.300e+03 +	2.300e+03 +	2.319e+03 +	2.300e+03 +	2.569e+03 +	2.300e+03
	std	7.826e+01	2.997e–01	4.828e–06	6.467e+01	7.872e+01	1.575e–02	1.373e–02	8.180e+01	1.901e–01	1.638e+02	8.890e–05
f_{10}	mean	2.599e+03 ≈	2.611e+03 +	2.605e+03 +	2.626e+03 +	2.599e+03 ≈	2.599e+03 ≈	2.599e+03 ≈	2.623e+03 +	2.618e+03 +	2.641e+03 +	2.599e+03
	std	1.741e–01	3.342e+01	2.536e+01	8.048e+00	2.502e–01	5.313e–01	2.582e–01	4.277e+01	3.974e+01	6.389e+01	3.343e–01
f_{11}	mean	2.600e+03 ≈	2.600e+03 +	2.600e+03 –	2.633e+03 +	2.600e+03 ≈	2.600e+03 +	2.600e+03 +	2.600e+03 +	2.600e+03 +	2.718e+03 +	2.600e+03
	std	8.181e–05	2.389e–02	1.047e–06	4.300e+00	1.089e–05	1.511e–03	4.586e–03	9.732e–03	1.740e–02	2.977e+02	3.875e–05
f_{12}	mean	2.866e+03 –	2.867e+03 +	2.865e+03 –	2.870e+03 +	2.866e+03 –	2.866e+03 ≈	2.866e+03 ≈	2.867e+03 ≈	2.867e+03 ≈	2.869e+03 ≈	2.867e+03
	std	2.123e+00	5.744e–01	5.078e–01	8.857e–01	8.858e–01	8.175e–01	5.602e–01	9.287e–01	5.397e–01	6.190e+00	2.800e–01
+/≈/–;		5/3/4	12/0/0	4/0/8	12/0/0	4/4/4	8/2/2	9/3/0	11/1/0	11/1/0	6/3/3	–
Avg. ranks:		4.8	8.2	2.7	10.7	3	5.4	5.5	6.9	7.0	6.7	3.8
Corrected ranks:		4	10	1	11	3	5	6	8	9	7	2

Table 25 The detailed experimental results in 20-D CEC2022

Func.		L-SHADE	CSO	iL-SHADE	MPEDA	jSO	L-SHADE-cnEpSin	MCSO	OLCSO	TPCSO	DEA ² H ²	SACSO
f_1	mean	1.616e+03 +	3.003e+02 +	3.000e+02 +	1.319e+04 +	3.000e+02 +	3.009e+02 +	3.000e+02 +	3.000e+02 +	3.000e+02 +	3.000e+02 +	$\approx 3.000e+02$
	std	5.738e+03	1.589e−01	1.488e−06	3.007e+03	3.897e−03	1.554e+00	3.267e−04	4.841e−03	1.776e−02	1.677e−07	3.361e−07
f_2	mean	4.863e+02 +	4.523e+02 $\approx$	4.489e+02 −	5.051e+02 +	4.492e+02 $\approx$	4.490e+02 $\approx$	4.492e+02 $\approx$	4.557e+02 +	4.538e+02 +	4.449e+02 $\approx$	$\approx 4.517e+02$
	std	1.623e+02	7.169e+00	9.132e−01	1.524e+01	5.307e−02	9.321e−01	4.451e−01	8.866e+00	8.770e+00	2.008e+01	5.362e+00
f_3	mean	6.000e+02 +	6.000e+02 +	6.000e+02 +	6.000e+02 +	6.000e+02 +	6.000e+02 +	6.000e+02 +	6.000e+02 +	6.000e+02 +	6.000e+02 +	6.000e+02
	std	2.418e−10	7.294e−09	7.275e−13	3.541e−03	7.669e−12	6.006e−08	1.554e−09	2.912e−11	1.157e−09	8.531e−12	4.687e−13
f_4	mean	8.024e+02 +	8.004e+02 +	8.017e+02 +	8.040e+02 +	8.028e+02 +	8.020e+02 +	8.004e+02 +	8.004e+02 +	8.003e+02 $\approx$	8.009e+02 +	8.003e+02
	std	1.317e+00	1.159e−01	3.493e−01	3.262e−01	3.628e−01	3.344e−01	1.197e−01	1.168e−01	8.677e−02	5.333e−01	8.498e−02
f_5	mean	9.000e+02 +	9.000e+02 +	9.000e+02 −	9.022e+02 +	9.000e+02 $\approx$	9.000e+02 +	9.000e+02 $\approx$	9.000e+02 $\approx$	9.000e+02 +	9.040e+02 +	9.000e+02
	std	1.849e−05	1.613e−05	4.476e−07	5.334e−01	2.672e−05	5.059e−04	1.979e−06	2.142e−05	1.960e−05	3.307e+00	3.646e−06
f_6	mean	5.466e+07 +	8.526e+04 +	1.153e+04 −	1.207e+08 +	3.232e+04 +	3.118e+04 +	3.059e+04 +	2.751e+04 $\approx$	3.735e+04 +	4.559e+04 +	2.506e+04
	std	1.320e+08	3.407e+04	5.037e+03	3.458e+07	6.971e+03	1.018e+04	6.978e+03	5.250e+03	8.904e+03	1.889e+04	4.070e+03
f_7	mean	2.100e+03 +	2.036e+03 +	2.034e+03 +	2.561e+03 +	2.049e+03 +	2.048e+03 +	2.030e+03 +	2.028e+03 +	2.034e+03 +	2.046e+03 +	2.023e+03
	std	2.791e+02	2.163e+00	2.884e+00	1.351e+02	6.241e+00	8.556e+00	3.965e+00	2.023e+00	2.452e+00	2.448e+01	1.097e+00
f_8	mean	2.673e+05 +	3.579e+03 +	2.227e+03 −	1.722e+05 +	2.253e+03 −	2.236e+03 −	2.877e+03 $\approx$	3.357e+03 +	3.354e+03 +	2.258e+03 −	2.884e+03
	std	1.155e+06	4.894e+02	4.145e+00	3.450e+05	2.113e+01	9.484e+00	3.360e+02	2.458e+02	4.576e+02	9.641e+01	2.204e+02
f_9	mean	2.690e+03 +	2.651e+03 +	2.636e+03 −	2.737e+03 +	2.637e+03 −	2.638e+03 −	2.637e+03 −	2.660e+03 +	2.651e+03 +	2.636e+03 −	2.647e+03
	std	1.152e+02	4.678e+00	1.241e−02	1.906e+01	7.199e−01	1.465e+00	6.262e−01	4.944e+00	5.471e+00	2.762e−02	2.295e+00
f_{10}	mean	2.809e+03 $\approx$	2.787e+03 $\approx$	2.764e+03 −	3.008e+03 +	2.759e+03 −	2.762e+03 −	2.764e+03 $\approx$	2.813e+03 +	2.793e+03 $\approx$	3.627e+03 +	2.793e+03
	std	1.382e+02	4.752e+01	2.410e+01	1.650e+02	1.594e+00	1.091e+00	2.525e+01	5.852e+01	5.162e+01	1.127e+03	5.028e+01
f_{11}	mean	2.662e+03 +	2.600e+03 +	2.600e+03 +	2.631e+03 +	2.600e+03 +	2.600e+03 +	2.600e+03 +	2.600e+03 +	2.600e+03 +	2.602e+03 +	2.600e+03
	std	2.358e+02	5.001e−03	5.355e−05	4.130e+00	5.719e−04	5.470e−02	7.371e−04	7.967e−01	5.120e−03	6.424e+00	3.090e−05
f_{12}	mean	2.950e+03 $\approx$	2.948e+03 $\approx$	2.943e+03 −	2.968e+03 +	2.940e+03 −	2.944e+03 −	2.946e+03 $\approx$	2.953e+03 $\approx$	2.950e+03 $\approx$	2.983e+03 +	2.952e+03
	std	1.900e+01	7.469e+00	1.244e+01	5.027e+00	1.994e+00	3.840e+00	6.569e+00	9.682e+00	6.122e+00	3.631e+01	9.180e+00
+/−/−;		10/2/0	9/3/0	5/0/7	12/0/0	6/2/4	7/1/4	6/5/1	10/2/0	8/4/0	8/3/1	−
Avg. ranks;		8.8	6.9	2.6	10.5	4.5	5.9	4.2	6.4	6.4	6.1	3.5
Corrected ranks;		10	9	1	11	4	5	3	7	7	6	2

Author Contributions All the authors contributed equally in this work.

Funding Not applicable

Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

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